



NACDeC-VI

(2022-23)

Sixth National Aerospace Conceptual Design Competition

Task -II Report

Submitted by

Collage Name	Chandigarh University
Team Name	AeroIndians
Team Leader Name	Venkat KSE
Team Member 2	Vinoth kumar G
Team Member 3	Prince Kumar Verma
Team Member 4	Gurleen Singh
Team Member 5	Gourab Koley

Under the guidance of

Dr.Gurmail Singh Malhi



**CHANDIGARH
UNIVERSITY**

Discover. Learn. Empower.

Certificate

Certified that this Task- I Review Report of Team: AeroIndians is approved by me for submission. Certified further that, to the best of my knowledge, the report represents work carried out by the students.

Date: 31/01/23

Signature

A handwritten signature in black ink, appearing to read 'Gurmail Singh Malhi', written diagonally.

**Dr. Gurmail Singh Malhi
(Faculty Mentor)**

Table of content

Abstract

1) Introduction

2) Structural Design of Aircraft

- 1. Discussion on Air Ambulance Design**
- 2. Basic three dimensional design of our Air Ambulance**
- 3. Canard Configuration and Downwash:**
- 4. Advantages over other types of VTOL and helicopters**
- 5. The concept of operations for medical emergencies using the designed vehicle.**
- 6. A Drawing of the Basic Arrangements of Paramedics and Patient**
- 7. Aerofoil Selection**
- 8. Drag Estimation**

3) Propulsion system of the Aircraft

- 1. Ducted Fan**
- 2. Advantages of our propulsion system**
- 3. Methodology and Calculations of Engine**
- 4. CFD Analysis of the Engine**
- 5. Distributed propulsion system**

4) Aircraft Sizing and Weight Estimations

- 1. Weight Estimation**
- 2. Sizing of the Aircraft**

5) Fuselage Design

- 1. Fuselage**
- 2. Types of Fuselages**

6) Wing Configuration

- 1. Geometry of the wing**
- 2. Degree of Freedom**

3. Trade of analysis with backup Reasoning

7) Landing Gear

8)Energy Distribution and Battery System

- 1. Battery technology and its energy consumption**
- 2. Schematic Layout of battery system**
- 3. Formulation and Calculations with Graphs**

9) Stability and Control

10) Communication system

- 1. Working Principle**
- 2. Line Diagram**

10) Reference



List Of Figures :

Figure 1.1 3D Cad model

Figure 1.2 Cad model of conceptual Design

Figure 1.3 Back View

Figure 1.4 The location of the door on the backside

Figure 1.5 The location of the door on the backside

Figure 1.6 Basic Compartment Division and arrangement

Figure 1.7 Cross Section View (Top view)

Figure 1.8 Cross Section View (Left view)

Figure 1.9 Cross Section View (Right view)

Figure 1.10 Cross Section Isometric View (NW)

Figure 1.11 Cross Section Isometric View (NE):

Figure 1.12 Wing

Figure 1.13 C_L v C_d graph

Figure 1.14 C_L v α graph

Figure 1.15 C_m v α graph

Figure 1.16 Airfoil Graph

Figure 2.1 over the free propeller

Figure 2.2 Disc Loading

Figure 2.3 2 Dimensional Representation

Figure 2.4 Cross Section Isometric View (NW)

Figure 2.5 Cross Section View (Right view)

Figure 2.6 Cross Section View (Front view)

Figure 2.7 Cross Section View (Back view)

Figure 2.8 Thrust /AMP/RPM/Speed/ef Graph

Figure:3.1- Pie Chart of the Weight Distribution of the Aircraft

Figure: 3.2- CAD design of the Aircraft

Figure: 3.3- Different Wing Configurations

Figure: 3.4- Tandem wing aircraft hypothesized advantages with respect to counteracting turbulence that helps our aircraft cruise at different situations

Figure: 4.1- Chord of the Wing

Figure 5.1 Hydrogen Fuel Cell

Figure 5.2 Schematic Layout of battery system

Figure 5.3

Figure 5.4 Torque(Nm) vs Speed (RPM)

Figure 5.5 Fan Pressure Ratio vs Corr. Mass Flow

Figure 5.6 Matlab simulation

Figure 6.1 Flight path

Figure 6.2 showing stability in conventional aircraft and its degrees of freedom.

Figure 6.3 showing the Streamline at the recirculation region attached on the upstream airfoil near stall

Figure 6.4 showing the velocity streamline for the tandem configuration

Figure 6.5 showing the use of the flap control for thrust vectoring

Fig 6.6 Similar feature used in our aircraft to assist for Vertical take - off and landing.

Figure 6.7 Cfd result depicting similar aircraft using thrust vectoring.

Abstract :

This task- II review a report for the conceptual Design of Air ambulance and in conjunction with the objectives mentioned in Task 1. The problems identified in the initial report have been briefed in detail. The Conceptual Design is shown in the report with 2D and 3D CAD diagrams with Selection of materials and airfoil . The Flight profile parameters have been calculated through empirical formulas. Sizing parameters have also been calculated. The Ducted Propulsion system combined with green energy initiated hydrogen fuel cell batteries to power the engines have been briefed in detail.

1. Introduction:

The concept of Air Ambulance is a revolutionary step in the history of aviation when compared to the conventional methods. This is one of the most efficient and fastest way for any rescue mission or any medical service. An air ambulance is a specially outfitted helicopter or fixed-wing aircraft that transports injured or sick people in a medical emergency or over distances or terrain impractical for a conventional ground ambulance. Fixed-wing aircraft are also more often used to move patients over long distances and for repatriation from foreign countries. These and related operations are called aeromedical. In some circumstances, the same aircraft may be used to search for missing or wanted people.

Like ground ambulances, air ambulances are equipped with medical equipment vital to monitoring and treating injured or ill patients. Common equipment for air ambulances includes medications, ventilators, ECGs and monitoring units, CPR equipment, and stretchers. A medically staffed and equipped air ambulance provides medical care in flight—while a non-medically equipped and staffed aircraft simply transports patients without care in flight. Military organizations and NATO refer to the former as medical evacuation (MEDEVAC) and to the latter as casualty evacuation (CASEVAC).

Air Traffic Control (ATC) grants special treatment to air ambulance operations, much like a ground ambulance using lights and a siren, only when they are actively operating with a patient. When this happens, air ambulance aircraft take the call sign MEDEVAC (formerly LIFEGUARD) and receive priority handling in the air and on the ground.

2. Structural Design of Aircraft :

2.1 Discussion on Air Ambulance Design:

After considering the criteria and range of operation, We came up with a design that has two wings, the main wing at the back and the small wing in front (canard). This design can take various forms, such as a tiltrotor, where the wing rotates between a vertical and horizontal position, or a tiltwing, where the entire wing tilts between a vertical and horizontal position. In both cases, the engines and propulsion system are typically located within the wing, and may be able to rotate or pivot as well to provide thrust in both the vertical and horizontal modes of flight. In contrast to these existing designs, we came up with a new one.

In our design, each wing is divided into two parts, front and back (like-aileron). Duct Fans attached to the aft portion of the wing will rotate when the aft part of the wing is tilted and the front part of the wing is fixed blended with the wing. A streamlined shape will reduce drag and improve performance. We are not yet finished with this design. In order to derive a good aerodynamic VTOL will have a balance of these characteristics optimized for its specific mission and operational requirements. The design is currently being analyzed using CFD to understand the aerodynamics and improve it further. The design of other subcomponents has been shown separately, and we will integrate them in the following tasks. A later part of the sections will cover the configuration and design of the propulsion system which will be incorporated with this design.

Winglets will be added to this wing section in order to improve aerodynamics and they help to reduce drag by controlling the flow of air around the wingtips. The design of components with vortex generators, and laminar flow surfaces to reduce drag and improve lift. Vortex generators are placed on the surface of the wings, tail, and other parts of the vehicle. They create small vortices of air that help to improve lift and reduce drag. Laminar flow surfaces are special surfaces that are designed to promote laminar flow, a type of airflow that is smooth and has less drag. This can be achieved through the use of special coatings, textures, or shapes.

2.2 Basic three dimensional design of our Air Ambulance: (complex design will be added in next steps)



Figure 2.1 Cad model

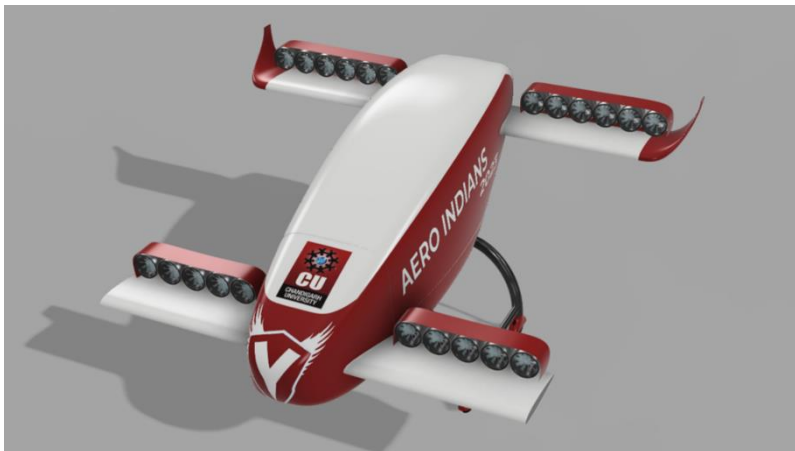


Figure 2.2 Cad model (Isometric View)



Figure 2.3 Cad model (Side View)



Figure 2.4 Cad model (Front View)



Figure 2.5 Cad model (Back View)

2.3 Canard Configuration and Downwash:

A canard is an aerodynamic configuration where the main wing is located at the back of the aircraft, while smaller wings (canards) are located at the front. When the canards generate lift, they also create a downward force, known as downwash. This downwash then impinges upon the main wing, which can affect the aircraft's lift and stability. This configuration can provide improved maneuverability and stability, but it also requires a more complex control system to manage the interaction between the canards and main wing.

One way to overcome downwash is by using a wing-to-wing configuration, where the main wing is placed behind the canard, creating less interference and thus reducing downwash. Another method is to use a slat or flap system on the

canard, which can be adjusted to change the angle of attack and reduce downwash. Another option is using a V-tail configuration, which reduces interference between the main wing and tail surfaces, again reducing downwash.

Longitudinal stability refers to the stability of an aircraft in relation to its pitch axis, which runs horizontally through the centre of the aircraft from nose to tail. A canard configuration can improve longitudinal stability by providing additional lift at the front of the aircraft, which helps to counteract the tendency of the aircraft to pitch nose-up or nose-down. Additionally, the canard can be used to control the angle of attack of the main wing, which can also help to improve longitudinal stability. Overall, a well-designed canard can make an aircraft more resistant to pitch-up or pitch-down movements, and therefore more stable in flight.

2.4 This design with a half tilt wing with attached duct fans has several advantages over other types of VTOL and helicopters. These include:

1. **High speed:** It can fly at much higher speeds than helicopters, making them well-suited for long-distance missions.
2. **Long range:** It has a much longer range than helicopters, allowing them to travel further without needing to refuel.
3. **Increased payload capacity:** This configuration can carry much heavier loads than helicopters, making them well-suited for air ambulances having patients and paramedics that require a lot of life supporting equipment.
4. **Increased safety:** Tiltrotors have a number of safety features that make them less likely to crash than helicopters. For example, they can fly at higher altitudes and in poor weather conditions, and they have the ability to continue flying even if one engine fails.
5. **Lower noise:** It generally produces less noise than helicopters and complete tilt wing VTOL, making them less disruptive to patient and operational areas like hospitals.

2.5 The concept of operations for medical emergencies using the designed vehicle.

The location of the door on the backside is as follows:

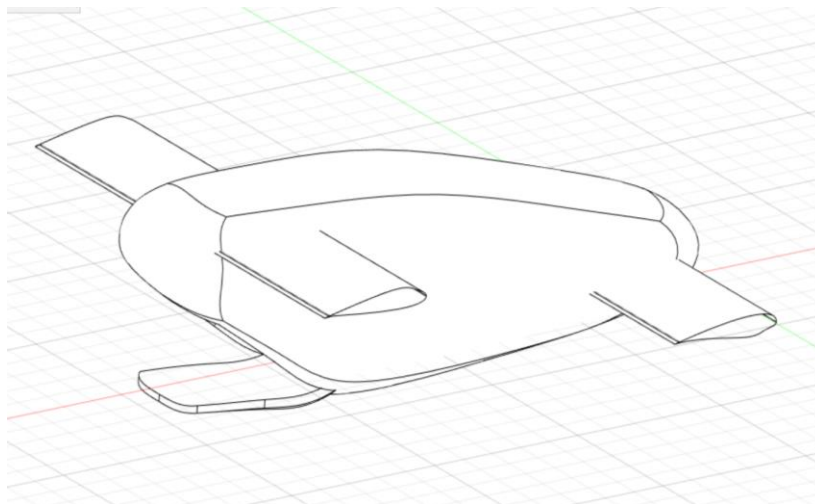


Figure 2.6

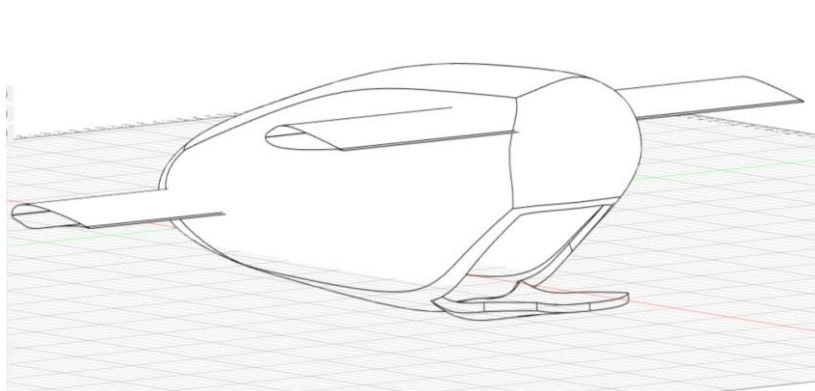


Figure 2.7

In this air ambulance design, the position of the door on the back side is used for patient loading and unloading. This allows for easy access to the patient compartment while the vtol is on the ground. Additionally, the back door location may provide more space for medical equipment and personnel inside the aircraft. The back door location of an air ambulance allows for easy access to the patient compartment while the vtol is on the ground. This is important because it allows medical personnel to quickly and safely load and unload patients who may be critically ill or injured. Additionally, this location may provide more space inside the aircraft for medical equipment, such as stretchers and monitoring equipment, as well as room for medical personnel to work on the patient during the mission.

One major advantage of having the door at the back is that it allows the patient to be loaded and unloaded directly into the aircraft without having to move them through tight spaces or around corners. This reduces the risk of further injury or complications during the process. Another advantage is that it allows the medical team to work in a more comfortable position. They have more space to work on the patient, and they can access the patient more easily without having to

move around the aircraft. Overall, after some trial and errors we came up with this position of the door on the back side and it is designed to provide easy access for patients and medical personnel, as well as to maximize the amount of space available for medical equipment and treatment.

The vehicle can accommodate a patient, paramedic, stretcher, and other medical equipment as follows:

Basic Idea for Compartment Division and Arrangements

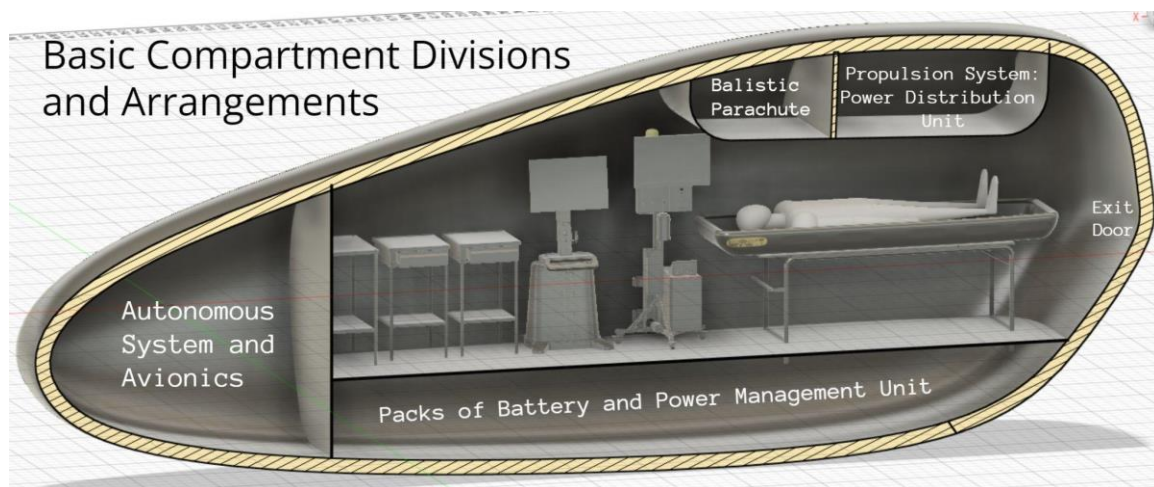


Figure 2.8

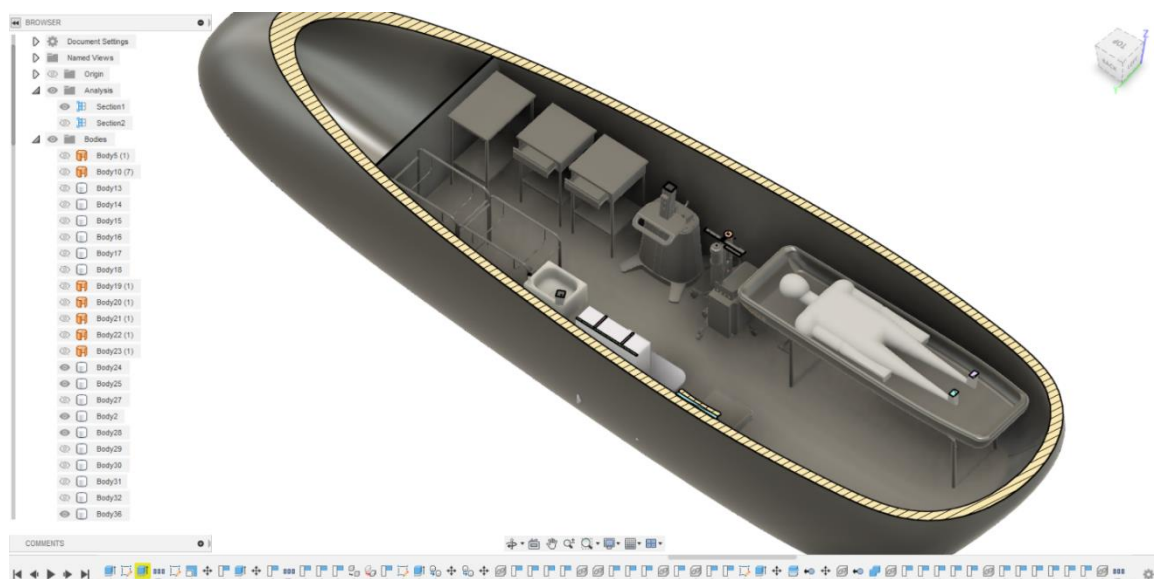


Figure 2.9

In terms of autonomous operation, this air ambulance needs advanced avionics, navigation and communication systems to ensure safe and efficient flight. This may include GPS navigation, weather radar, and a traffic collision avoidance system (TCAS). In order to perform this operation, advanced computers are required. A frontal part of this vehicle will have these systems placed to make it easier for the sensors and navigation systems to operate.

The importance of ballistic parachute assisted in this design is that the canard in front of the main wing in a certain height difference makes it difficult to safely land the vehicle during unfortunate situations like power cuts or malfunctions of the power unit. So, we need to include the ballistic parachute in this design to provide safety and handle difficult situations. It will be activated in such circumstances and help in safely landing the air ambulance. It will be approximated to the location of the center of gravity for level rescue.

Power distribution units (PDU) are important in electric VTOL that distribute electrical power throughout the aircraft. In an air ambulance, the PDU is responsible for providing power to the various systems and equipment on board, including the avionics, medical equipment, and lighting. The PDU may also include backup power systems in case of a main power failure.

Compartments will include:

1. **Main electrical panel:** This is the main control center for the PDU, where the pilot or medical personnel can monitor and control the distribution of electrical power.
2. **Power inverters:** These convert direct current (DC) power from the aircraft's batteries into alternating current (AC) power for use by the various systems and equipment.
3. **Circuit breaker panels:** These protect the aircraft's electrical systems by automatically shutting off power in the event of an overload or short circuit.
4. **Backup power systems:** These provide emergency power in the event of a main power failure. These can be batteries or generators.
5. **Power distribution buses:** These distribute power from the main electrical panel to the various systems and equipment throughout the aircraft.
6. **Voltage regulators:** These ensure that the voltage level remains consistent, to protect the equipment from damage due to voltage fluctuations.

2.6 A Drawing of the Basic Arrangements of Paramedics and Patient:

Cross Section View (Top view):

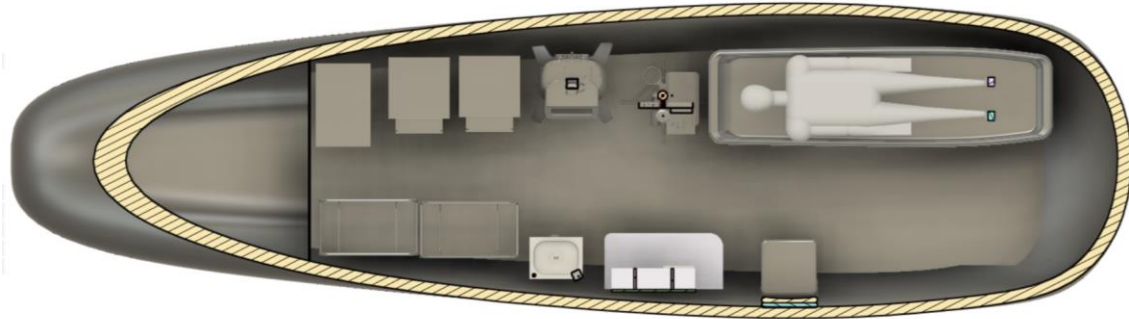


Figure 2.10

Cross Section View (Left view):

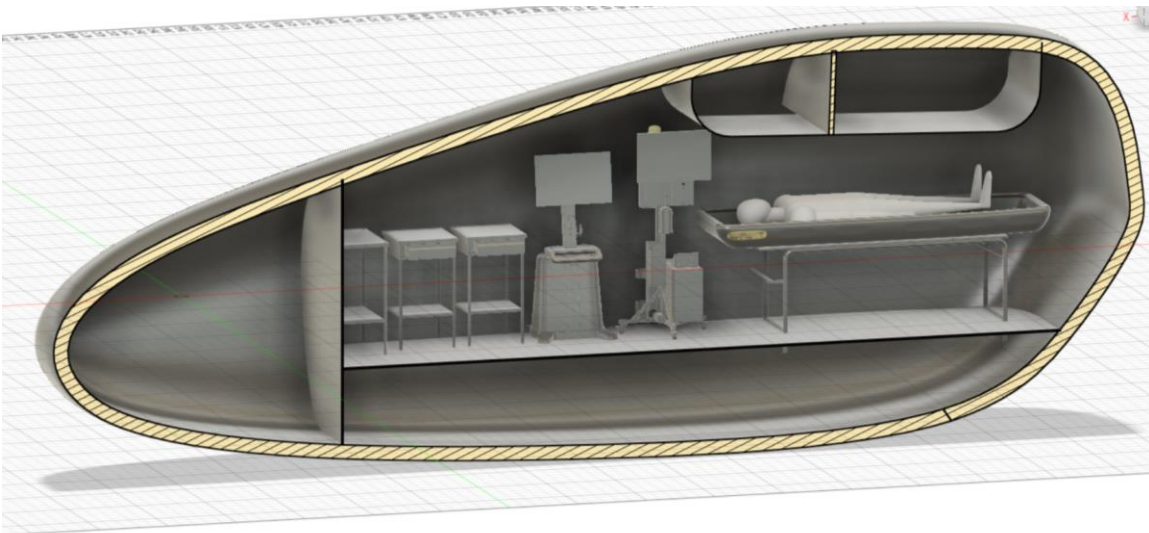


Figure 2.11

Cross Section View (Right view):

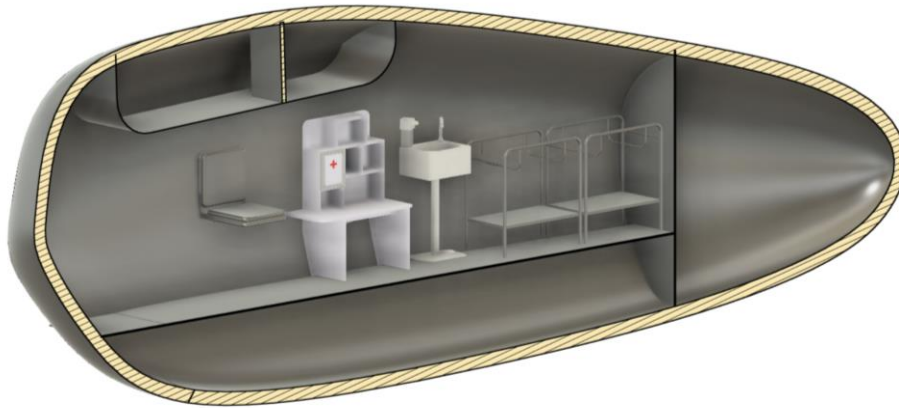


Figure 2.12

Cross Section Isometric View (NW):

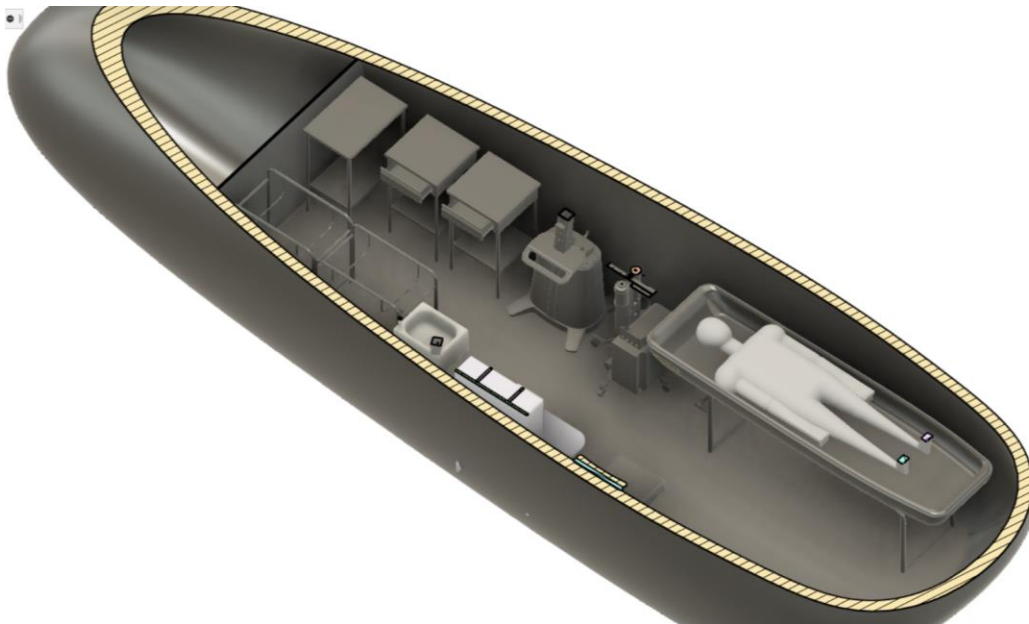


Figure 2.13

Cross Section Isometric View (NE):

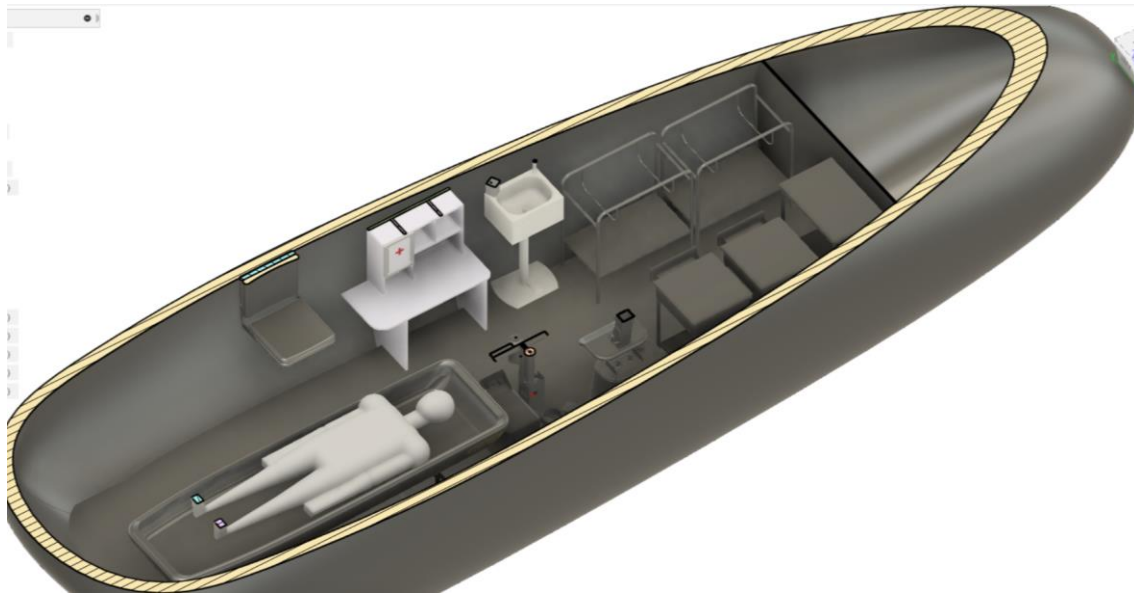


Figure 2.14

In this task, we present the basic design of an air ambulance interior and arrangement of paramedics and other equipment. In further tasks, the design will be optimised for the safe and efficient transport of critically ill or injured patients by making wall attached or floor attached equipment. The final design includes a patient compartment with a stretcher, monitoring equipment, oxygen supply, and storage for medical supplies.

To make the structure more robust, each component and attachment will be extruded from a frame, and a supporting layer with some thickness will be provided. The medical crew also have space to work and provide care during transport missions. Additionally, advanced equipment such as a ventilator or defibrillator have been taken into consideration.

We need to equip the air ambulance with an electric stretcher which offers several advantages over a manual stretcher. One main advantage is that it allows for easy and efficient loading and unloading of the patient. Electric stretchers can be raised and lowered to the ground using a remote control, eliminating the need for manual lifting and reducing the risk of injury to the medical staff. Additionally, electric stretchers can be adjusted to different positions, such as a seated position, which can provide added comfort and support for the patient during transport. Another advantage is that electric stretchers can be used to move patients in tight spaces, such as small aircrafts or helicopters, where a manual stretcher may not

fit. Overall, electric stretchers can improve patient care and safety, as well as the efficiency of air ambulance operations.

Electric stretchers for air ambulances typically have several features that enhance patient care and safety during transport. Some of these features include:

1. **Adjustable height:** The stretcher can be raised and lowered to different heights using a remote control, which makes it easier to load and unload the patient from the ambulance or aircraft.
2. **Multiple positions:** Electric stretchers can be adjusted to different positions, such as a seated position, which can provide added comfort and support for the patient during transport. This can be particularly useful for patients with spinal injuries or other conditions that require a specific position.
3. **Power drive:** Electric stretchers have an electric drive system that allows the stretcher to be moved around smoothly and easily, even in tight spaces. This can help reduce the risk of injury to medical staff and patients.
4. **Stabilizing wheels:** Some electric stretchers have wheels that can be locked in place to keep the stretcher stable during transport.
5. **Patient restraint system:** Some electric stretchers have built-in restraint systems, such as straps or harnesses, to keep the patient securely in place during transport.
6. **Monitoring systems:** Some electric stretchers come with built-in monitoring systems, such as heart rate monitors or oxygen saturation monitors, that can provide vital information about the patient's condition during transport.

2.7 Aerofoil Selection:

2.7.1 Aerofoil

Aerofoil or Aerofoil is a cross-sectional shape designed with a curved surface, giving it the most favourable ratio between lift and drag in flight. Lift is the component such that the force is perpendicular to the direction of motion and drag is the component parallel to the direction of motion. A similar idea is used in designing hydrofoils, which is used when water is used as the working fluid. Aerofoils are highly efficient lifting shapes as they generate more lift than similarly sized flat plates of the same area and generate lift with significantly less drag.

The designing of the aerofoil depends on the aerodynamic characteristics, which further depends on the aircraft's weight, speed, and purpose. These are dependent on specific terms that need to be defined to understand the design. German

mathematician Max Munk first devised aerofoil, and it was further refined by British aerodynamicist Hermann Glauert and others in the 1920s.

2.7.2 Aerofoil Terminology:

Following are the terms that are related to aerofoils:

- **Chord:** A chord is defined as the distance between the leading edge, which is the point at the front of the aerofoil and has maximum curvature and the trailing edge which is the point at the rear of the aerofoil with maximum curvature along the chord line.
- **Chord line:** A chord line is defined as the straight line connecting the leading and trailing edges.
- **Upper surface:** The upper surface is also known as the suction surface, associated with high velocity and low static pressure.
- **Lower surface:** The lower surface is also known as the pressure surface with higher static pressure.
- **Aerodynamic centre:** The pitching moment is independent of lift coefficient and angle of attack (AOA) at this centre.
- **Centre of pressure:** The pitching moment is zero at this centre.
- **The angle of attack (AOA):** The angle formed between a reference line on a body and the oncoming flow.
- **Pitching moment:** The moment or torque produced the aerodynamic force on the aerofoil.

2.7.3 Types of Aerofoils:

Aerofoils are classified into two types as symmetrical and nonsymmetrical aerofoils.

- **Symmetrical Aerofoil:** This has identical upper and lower surfaces such that the chord line and mean camber line are the same producing no lift at zero AOA. These find applications in most of the light helicopters in their main rotor blades.
- **Non-symmetrical aerofoil:** It is also known as a cambered aerofoil. This has different upper and lower surfaces such that the chord line is placed above with large curvature. These have different chord line and camber line. The advantage of non-symmetrical aerofoil is that the lift to drag ratio and stall characteristics are better and useful lift is produced at zero AOA. The disadvantages are that they are not economical and there is a production of undesirable torque.

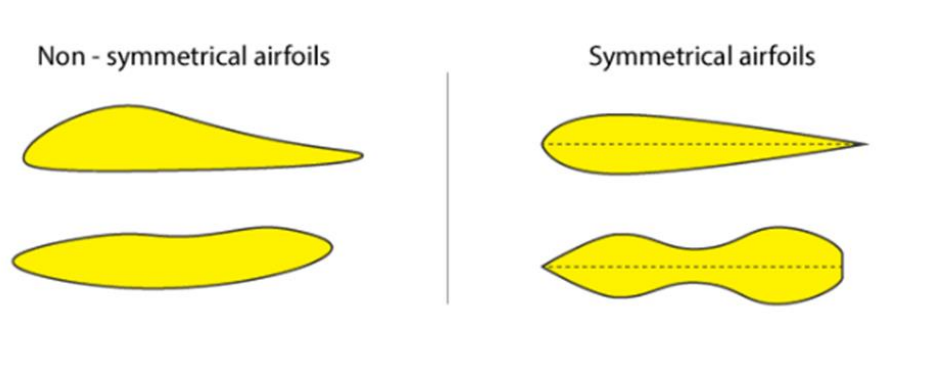


Fig- 2.15 (Types of Air foils)

2.7.4 Aerofoil Selection:

Aerofoil Selection is an iterative process through which we need to choose our desired aerofoil for the aircraft. The aerofoil is chosen based on speed factor, Aerofoil thickness to chord ratio of the aerofoil and the stall characteristic of the aerofoil.

Air foil thickness (t/c) has an impact on drag, maximum lift; stall characteristics, structural weight and critical Mach number ($t/c \sim 3 - 18 \%$)

aerofoil	Cl, max	AOA stall	t/c %	Cd, min	Re	Cm, ac	L/D
Naca 0015	1.4029	17	15	0.00632	100000	0.0359	77.87697
naca 4415	1.5425	17	15	0.00659	100000	-0.0145	119.3679
naca 23015	1.5586	17	15	0.00719	100000	0.0276	90.95238
naca 63015A	1.2465	16.5	15	0.0061	100000	0.0286	72.81768
naca 642215	1.3288	17	15	-1.0969	100000	0.0002	99.17127
jowgasky 0015	1.4485	16	15	0.00697	100000	0.0242	85.84821

Fig- 2.16 (6-Different Air foils)

And the required comparison is done based on Reynolds number, lift coefficient, angle of attack, etc. The comparison graph is shown below

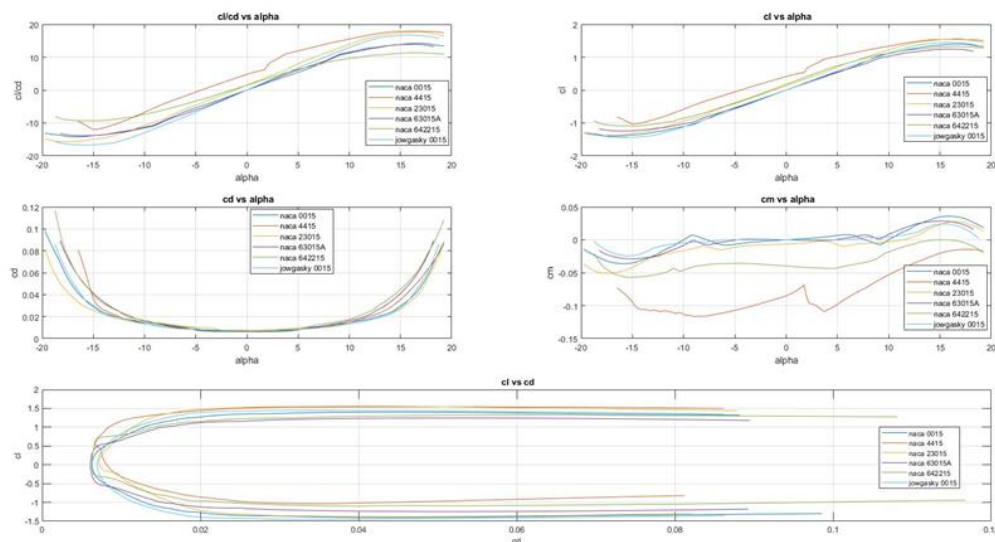


Fig- 2.17 Air foil Comparisons

According to the comparison we have chosen the aerofoil 'NACA 642215' because of its improved performance and especially due to its laminar flow over the wing surface.

NACA 642215 has the following characteristics:

1. Maximum camber: 6% of the chord length
2. Maximum camber position: 40% of the chord length
3. Maximum thickness: 21% of the chord length

The lift- coefficient of the aircraft is calculated by the formula $C_{L,c} = \frac{2(\frac{W}{S})}{(V_c)^2 * \rho}$ and the calculated values is found to be 0.0655. The lift- coefficient for the wing is then calculated by the equation.

$C_{L,c@wing} = \frac{C_{L,c}}{0.95}$ on which we have found the value to be 0.0106. The required lift- coefficient of the aerofoil is calculated by the formula, $C_{L,c@airfoil} = \frac{C_{L,c@wing}}{0.9}$ and it is found to be 0.0117.

The Aspect Ratio are then calculated by the equations.

$$AR = \frac{b^2}{S} = \frac{\text{span}}{\text{mean cord}} = \frac{\text{span}^2}{\text{wing area}}, \left(\frac{L}{D}\right)_{\max} = \frac{1}{2\sqrt{k * C_{D,0}}}$$

$$\text{where } k = \frac{1}{\pi * e * AR}$$

Where K is the wing induced drag factor, e is the Oswald span efficiency factor, and $C_{D,0}$ is the aircrafts zero-lift drag coefficient.

$$C_{D,i} = \frac{C_L^2}{\pi * e * AR}$$

From the initial sizing we got the data of wing loading $\frac{W}{S} = 89.47$. Considering the density at sea level to be 1.225 kg/m^3 . Assuming the $AR = 8$ and from the previous data point total weight $W_{TO} = 1300$. The wing area(S) and wingspan(b) are calculated and found to be 14.5300 m^2 and 10.7815 m respectively. The formula used here for wing area(S) is $S = \frac{W_{TO}}{(\frac{W}{S})}$ and the wingspan(b) $b = (AR * S)^{0.5}$

Similarly, for the calculation of tip cord, root cord and taper ratio. The taper ratio is 0.35. As taper ration lower than 0.2 should be avoided for all because it promotes tip stall. The taper ratio is the ratio between the tip chord and root chord (varies between 0 to 1).

$$\lambda = \frac{C_T}{C_R}, \quad C_{\text{root}} = \frac{2S}{b(1+\lambda)} \quad \text{and} \quad C_{\text{tip}} = \lambda * C_{\text{root}}$$

From calculations the obtained values of root cord are 1.99966 m, and the tip cord is 0.6988 m.

The mean aerodynamic cord is calculated from the formula,

$$\hat{c} = \left(\frac{2}{3}\right) * C_{\text{root}} * \left(\frac{1+\lambda+\lambda^2}{1+\lambda}\right) \quad \text{and it is found to be } 1.4518 \text{ m.}$$

The sweep angle of a wing is between a line drawn along the span at a constant fraction of the chord from the leading edge, and a line perpendicular to the centreline. The sweep angle is found by, $\tan \Lambda_{\frac{c}{4}} = \tan \Lambda_{LE} + \frac{C_{root}}{2b} (\lambda - 1)$ and it is calculated to be 3.4442 degrees.

According to the Raymers textbook, the wing incidence are generally 2-4 degrees for GA aircraft. While for the Dihedral angle

- a. Low wing -> tends to have considerably greater dihedral angle than high wing
- b. High wing -> Gives negative dihedral -> anhedral to balance b/w lateral stability and roll control
- c.

Wing types	Angles
High – wings	0 – 1
Mid – wings	2 – 4
Low – wings	3 – 6

After calculations and different iterative studies, we have obtained the required dimensions of wings with the 'NACA 642215' as the required aerofoil for the best performance.

2.8 Drag Estimation:

What is drag?

Drag is the aerodynamic force that opposes an aircraft's motion through the air. Drag is generated by every part of the airplane. Drag is a **mechanical force**. It is generated by the interaction and contact of a solid body with a **fluid** (liquid or gas). It is not generated by a force field, in the sense of a gravitational field or an **electromagnetic field**, where one object can affect another object without being in physical contact. For drag to be generated, the solid body must be in contact with the fluid. If there is no fluid, there is no drag. Drag is generated by the difference in velocity between the solid object and the fluid. There must be motion between the object and the fluid. If there is no motion, there is no drag. It makes no difference whether the object moves through a static fluid or whether the fluid moves past a static solid object.

Drag Force Example

An Airplane Flying



Science Facts .net

Fig- 2.18 Drag Force

Drag Force Example

An Airplane Flying

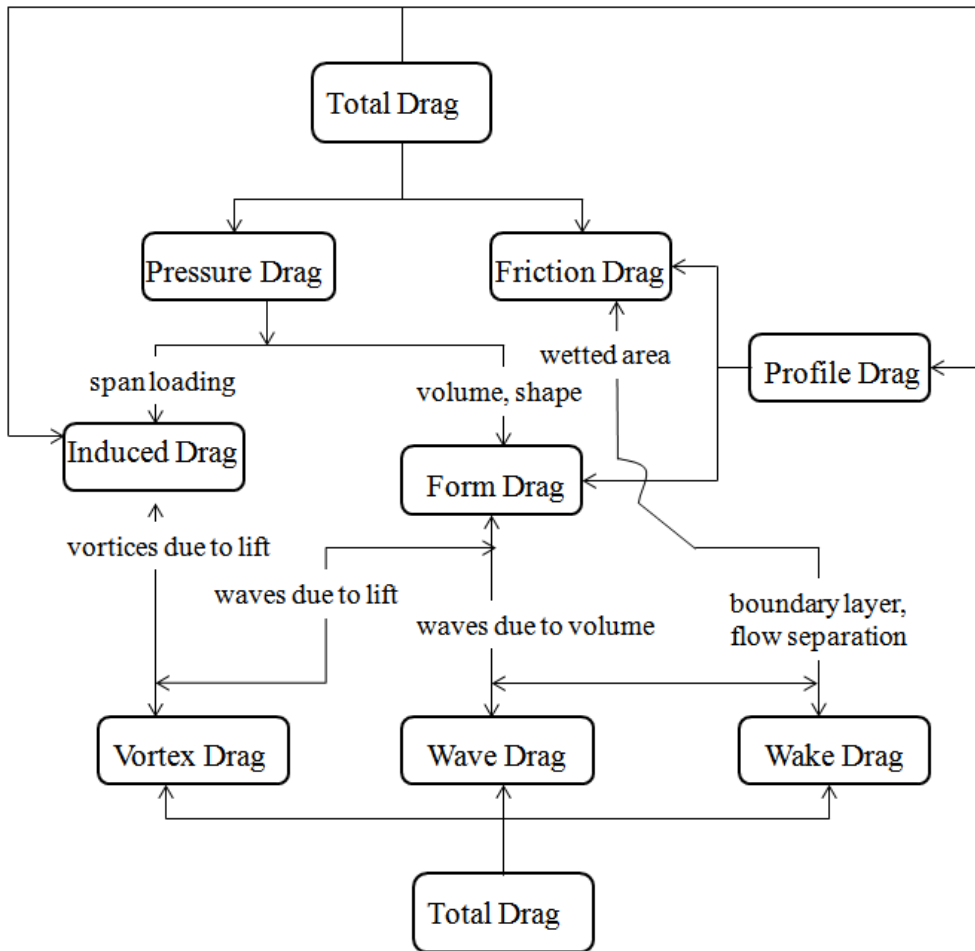


Science Facts .net

Types of drag:

1. Parasitic drag
2. Induced drag.

3. Wave drag.



Parasite drag:

parasitic drag = skin friction drag + form drag + interference drag

Estimation of Total drag co-efficient

$$C_D = C_{D,0} + k * C_L^2 + C_{D,W}$$

where

$C_{D,0}$ = Zero – Lift drag coefficient

k = Lift – induced drag coefficient factor

C_L = Lift Coefficient

$C_{D,W}$ = Wave drag coefficient

Where $k = \frac{1}{\pi * e * AR}$

$$C_{D,0} = (C_{D,0})_{\text{clean}} + (C_{D,0})_{\text{HLD}} + (C_{D,0})_{\text{L.G}}$$

HLD = high lifting device

@cruise,

$$C_{D,0} = (C_{D,0})_{\text{clean}}$$

@Take-off & Landing

$$C_{D,0} = (C_{D,0})_{\text{clean}} + (\Delta C_{D,0})_{\text{HLD}} + (\Delta C_{D,0})_{\text{L.G}}$$

$$(C_{D,0})_{\text{clean}} = (C_{D,0})_{\text{components}} + (\Delta C_{D,0})_{\text{leakage}} + (\Delta C_{D,0})_{\text{miscell}}$$

Table: Leakage & perturbation Drag	
Aircraft Type	Drag %
Propeller aircraft	5- 10
Jet Transport/ Bomber	2- 5

Clean Zero Lift drag coefficient

$$(C_{D,0})_{\text{components}} = (C_{D,0})_{\text{wing}} + (C_{D,0})_{\text{fuselage}} + (C_{D,0})_{\text{HT}} + (C_{D,0})_{\text{VT}} + (C_{D,0})_{\text{nacelle}}$$

$$(C_{D,0})_{\text{component}} = \sum_{k=1}^{\text{components}} C_{f,e} * FF * Q * \frac{S_{\text{wet}}}{S_{\text{ref}}}$$

where $C_{f,e}$ = skin friction drag coefficient

FF = form factor

Q = Interference factor

S_{wet} = wetted surface area and S_{ref} = reference surface area

The parasitic drag coefficient of wing is calculated from the formula :

$$(C_{D,0})_{\text{wing}} = C_{f,e} * FF * Q * \frac{S_{\text{wet}}}{S_{\text{ref}}}$$

The skin friction drag coefficient is calculated from the formula

$$C_{f,\text{turbulent}} = \frac{0.455}{(\log Re)^{2.58} + (1 + 0.144 * M^2)^{0.65}}$$

And it is obtained as 4.03×10^{-04}

Reynold's number, $Re = \frac{V * l}{\nu}$

l = characteristics length, calculated as 4.16×10^6

Mean Aerodynamic Chord of Exposed wing and empennage. Fuselage length and nacelle length.

$$Re = \frac{V * C_{mac,e}}{v}$$

$$C_{mac,e} = \frac{2}{3} C_{R,e} \left(\frac{1 + \lambda_e + \lambda_e^2}{1 + \lambda_e} \right)$$

$$C_{R,e} = C_R - \left(\frac{C_R - C_T}{\frac{b}{2}} * \frac{D_f}{2} \right)$$

$$\lambda_e = \frac{C_T}{C_{R,e}}$$

Skin Roughness values

Surface Type	K(C in ft)
Camouflage paint on aluminium	$3.33 * 10^{-5}$
Smooth paint	$2.08 * 10^{-5}$

$$K = 1.015 * 10^{-5} \text{ m}$$

Finding the cut-off Reynolds number

Subsonic: $Re_{cutoff} = 38.21 \left(\frac{C}{k} \right)^{1.053}$ where $\left(\frac{C}{k} \right) = \left(\frac{C_{mac,e}}{k} \right)$ and the cut off Reynolds number has been calculated as $1.0253 * 10^7$

The form factor is calculated from

$$FF = \left[1 + \frac{0.6}{\left(\frac{x}{c} \right)_m} \frac{t}{c} + 100 \left(\frac{t}{c} \right)^4 \right] \left[1.34 * M^{0.18} * \left(\cos \Lambda_{\frac{c}{4}} \right)^{0.28} \right] \text{ for sweep wing}$$

The foam factor is calculated as 1.6145

The Interference factor is taken from the table as given by Raymer

Table: Interference Factor		
Interference factor with respect to...	Property	Interference factor Q
Nacelle	Engine mounted directly on the wing of fuselage	1.5

	Distance of engine to wing respectively fuselage is smaller than the engine diameter d_N	1.3
	Distance of engine to wing respectively fuselage is greater than the engine diameter d_N	1.0
Wing	High wing, mid wing, or low wing position with aerodynamically wing-fuselage fairing	1.0
	Low-wing position without aerodynamically optimized wing fuselage fairing	1.101.40

Hence we are taking $Q = 1$

The wetted surface area for the wing is calculated from the formula

$$S_{\text{wet}} = 2 * S_{\text{exposed}} \left(1 + 0.25 \left(\frac{t}{c} \right)_R \frac{1 + \tau \lambda}{1 + \lambda} \right)$$

$$\text{Where } \tau = \frac{\left(\frac{t}{c} \right)_T}{\left(\frac{t}{c} \right)_R}$$

$$S_{\text{exposed}} = 2 * \left(\frac{b}{2} \right)_e \left(\frac{C_{R,e} + C_T}{2} \right)$$

$$\text{Where } \left(\frac{b}{2} \right)_e = \left(\frac{b}{2} \right) - \left(\frac{D_f}{2} \right)$$

The wetted reference area is calculated as 25.2914 m² .

The parasitic drag calculation of wings is 0.0011.

The parasitic drag of fuselage is same as that of the wing with slight modification

$$(C_{D,0})_{\text{wing}} = C_{f,e} * FF * Q * \frac{S_{\text{wet}}}{S_{\text{ref}}}$$

The skin friction drag coefficient is calculated from the formula.

$$C_{f,\text{turbulent}} = \frac{0.455}{(\log Re)^{2.58} + (1 + 0.144 * M^2)^{0.65}}$$

And it is obtained as $2.98 * 10^{-04}$

$$\text{Reynold's number, } Re = \frac{V * l}{\nu}$$

l = fuselage length , calculated as $4.16 * 10^6$

Mean Aerodynamic Chord of Exposed wing and empennage. Fuselage length and nacelle length.

$$Re = \frac{V * L_{\text{Fuselage}}}{\nu}$$

And it is calculated as $2.7612 * 10^7$

The cut- off Reynolds number is taken with the same formula, and it is obtained as $7.5143 * 10^7$

The foam factor is taken with the formula

$$FF_{\text{fuselage}} = 1 + \frac{60}{\left(\frac{l_F}{d_F}\right)^3} + \frac{\left(\frac{l_F}{d_F}\right)}{400} \text{ and it is obtained as } 1.3744$$

The interference factor of fuselage is taken as $Q = 1.0$

The wetted surface area is calculated as

$$S_{\text{wet}} = \pi D_f L_f \left(1 - \frac{2}{\left(\frac{L_f}{D_f}\right)} \right)^{\frac{2}{3}} \left(1 + \frac{1}{\left(\frac{L_f}{D_f}\right)^2} \right)$$

And it is value obtained from calculations is 40.4436 m^2 .

The calculated drag of fuselage is 0.0011

Similarly for the engine,

The length and diameter of the engine nacelle is obtained as 0.4387 and 0.395 m.

$Re = \frac{V * L_N}{\nu}$ where L_N = length of the nacelle and then the cut-off Reynolds number

Calculated as $1.258 * 10^6$ and $2.9078 * 10^6$ respectively.

The $C_{f,turbulent}$ is obtained as $4.97 * 10^{-04}$

The form factor for nacelle $FF_N = 1 + \left(\frac{0.35}{\left(\frac{L_N}{d_N}\right)}\right)$ and it is obtained as 1.351

The interference drag is taken as $Q = 1.5$ for engine mounted directly above the wing.

For the wetted surface area, $S_{wet} = 30 * 1.2 * \pi D_N L_N$, 30 is taken due to the number of engines and it is obtained as 19.5982

The total component drag is calculated by the drag over the wing, fuselage and engine nacelle. 0.0013 is obtained as the component drag after calculating and the drag over the horizontal and vertical tail is taken as 0 since our model lacks the concept of tail and we use thrust vectoring.

Considering 5% of the component drag as the leakage drag

$$(\Delta C_{D,0})_{leakage} = 5\% (C_{D,0})_{components}$$

Therefore, leakage drag = $1.7984 * 10^{-04}$

However, the miscellaneous drag is considered to be 0.00320 which consist the drag over the landing gear and the high lifting device .

The total parasitic drag is then calculated as the drag over the components, leakage and the miscellaneous drag added together. The total value is obtained as 0.0358.

The lift induced drag coefficient is calculated from the formula

$$C_D = C_{D,0} + K C_l^2$$

Where induced drag, $C_{D,i} = k C_L^2$ obtained as $4.356 * 10^{-8}$

The take-off drag is calculated with the formula $C_{L,TO} = \frac{2(\frac{W}{S})}{\rho * V_{TO}^2}$ and it is obtained as $1.20 * 10^{-5}$ which is required for the drag calculations.

$$C_{D,TO} = (C_{D,0})_{TO,l} + k C_{L,TO}^2$$

$$V_{TO} = 1.2 * V_{stall}$$

However $(C_{D,0})_{TO,l}$ is considered to be 0.25 as it's a vertical take- off and landing.

The value obtained thus is 274.692

The lift coefficient for landing is thus, $C_{L,L} = \frac{C_{L,TO}}{1.2}$ The value obtained thus is 0.9860.

And the drag coefficient during landing is $C_{D,L} = (C_{D,0})_{TO,l} + k C_{L,L}^2$

The value obtained thus is 560.3571 .

The drag co-efficient during cruise co-efficient is

$$C_D = C_{D,0} + K C_L^2$$

The value obtained thus is 0.0210.

The drag for cruise, take-off and landing are thus calculated from the formula

$$\text{Drag} = 0.5 * \rho * V^2 * S * C_D$$

With the respective changes in density and velocity at certain altitudes.

The value thus obtained is $\text{Drag}_{\text{cruise}} = 179.720 \text{ N}$.

$\text{Drag}_{\text{take-off}} = 274.692 \text{ N}$ and $\text{Drag}_{\text{landing}} = 560.3571 \text{ N}$.

3. Propulsion system of the Aircraft

3.1 Ducted Fan

Ducted fans or ducted propellers are promising devices for high static thrust propulsion systems. Ducted propellers typically produce more static thrust compared to independent propellers of the same diameter and power rating. Additionally, ducted fan systems offer an additional safety feature by housing the rotating fan inside the duct, making it an attractive option for a variety of advanced UAV configurations. Driven by advances in the field of electric propulsion systems, a number of new small aircraft and propulsion concepts have been developed in recent years. Hybrid and electric drives offer greater design freedom and enable new aircraft concepts. When installed on conventional aircraft, it can help reduce fuel consumption and emissions. Other secondary benefits include reduced noise and increased operational flexibility.

3.2 Advantages of our propulsion system

Noise reduction is very important to ensure that future aircrafts can be operated and accepted in urban environments. As its an important factor, therefore needs to be considered at the conceptual design stage only. This is where ducted fan offer a potential advantage over free propellers. This is due to the following: non-propagation of rotor alone noise outside the duct; a variety of rotor blade and stator vane design features usable to minimize noise; noise shielding options; possible use of acoustic liners; and a stronger absorption of higher frequency noise. All of the aforementioned reduce noise compared to the conventional unrestricted radial emission of propeller noise. Duct housings and acoustic liners contain noise and prevent it from radiating in all directions like an open propeller.

In this regard, the ducted fan offers potential advantages over the free propeller.

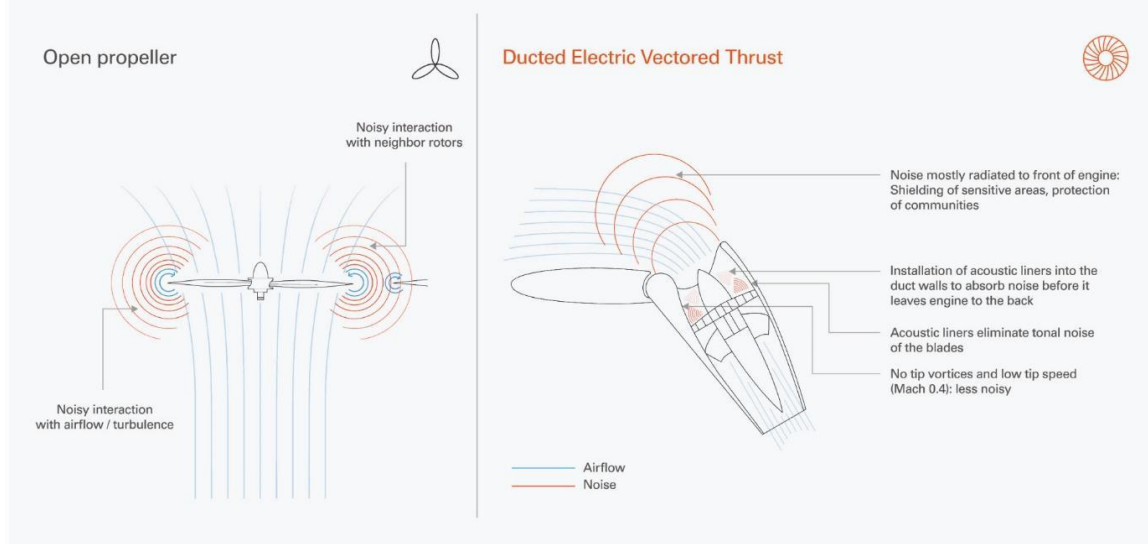


Figure 3.1

- **LINERS:** we have introduced acoustic liners in between the engine in such a way that it not even disturbs the airflow and also reduce the noise. Engine duct with Acoustic liners capture and decrease the noise before it leaves into environment. Duct housings and acoustic liners contain noise and prevent it from radiating in all directions like an open propeller.



Figure 3.2

- **TWIST IN BLADES:** as the tip of the blade travels more distance than at root, so to keep the airflow equivalent at all the points of blade the twist is different (more at root and less at the tip). As the blade is twisted air can't settle and create shockwaves.

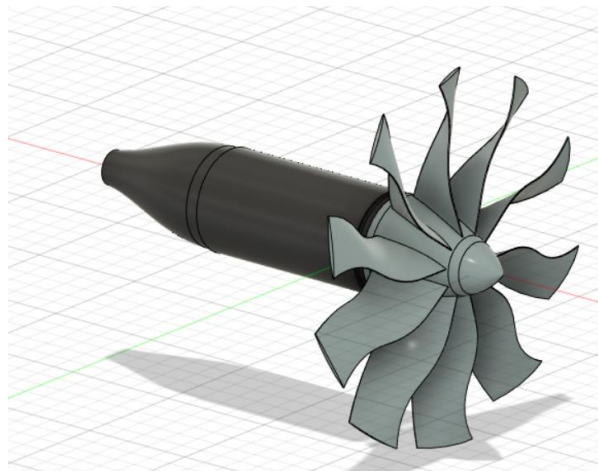


Figure 3.3

- **THRUST VECTORING:** Our aircraft can be controlled using Thrust vectoring without the need of standard control surfaces like ailerons, elevators and rudder. The engines only receive two signals: its speed and the angle of flap.

- **Variable nozzle:** The biggest challenge was designing a nozzle which is efficient at both cruise and hover state. So a variable nozzle at exit ensures peak efficiency in both cruise and hover modes. As the requirement of thrust is ten times in hover state as compared to in cruise, so at hover state we will increase the area of nozzle to increase efficiency.



Figure 3.4

Electric jet engines integrated into the wing flaps offer advantage in payload, aerodynamic efficiency, and low-noise profile, whilst air ambulances also provide thrust vector control for maneuvering through each phase of flight.

- Zero operational emissions
- Highly efficient cruise phase
- Vertical take-off and landing for urban accessibility
- Low noise for frequent urban operations
- Suitable for such flights a mandatory technical profile for air ambulances is the electric ducted fan. In hover, the ducted fan exhibits about 40% higher efficiency at the same disc load compared to the open propeller. This is because the presence of ducts and guide vanes eliminates nearly all blade tip losses and vortex losses. This efficiency gain somewhat compensates for the higher power requirements caused by the 10x disk load while hovering.

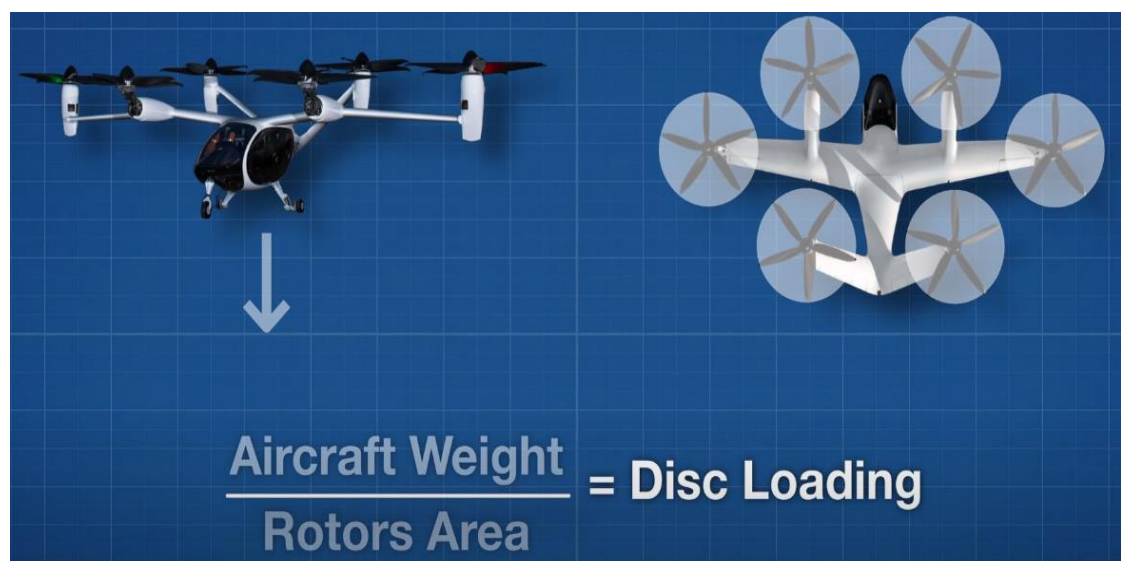


Figure 3.5

2 Dimensional Representation :

The below diagram Represents the dimensions and accurate measurement of the ducted engine that we have designed specifically for this Air Ambulance to act effectively as much as possible.

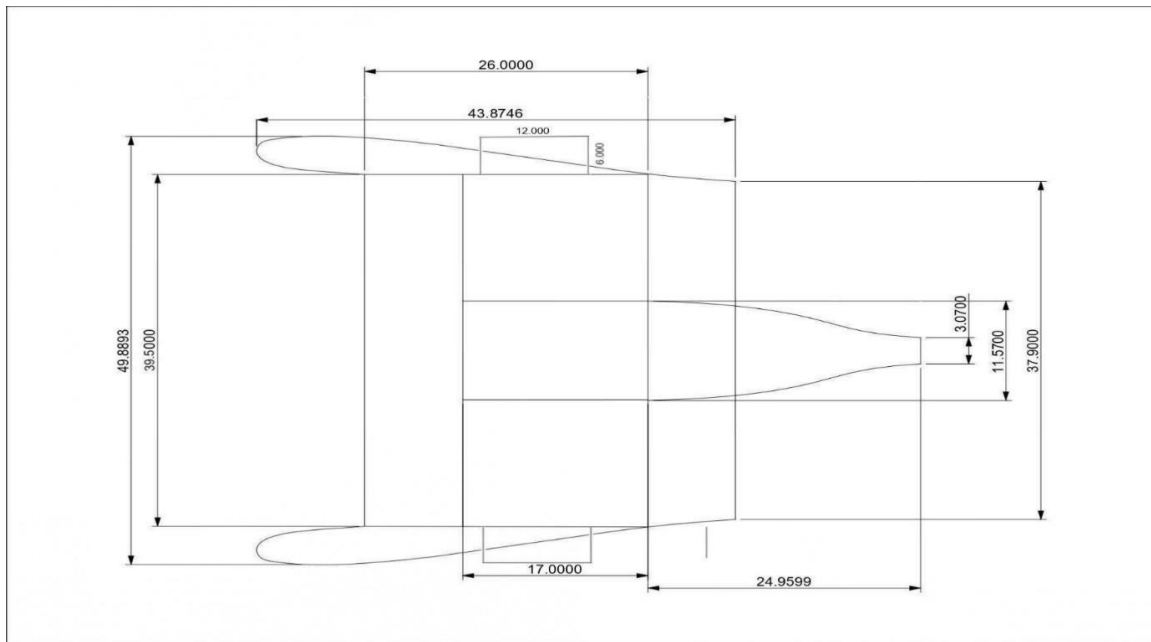


Figure 3.6

3 Dimensional CAD Design :

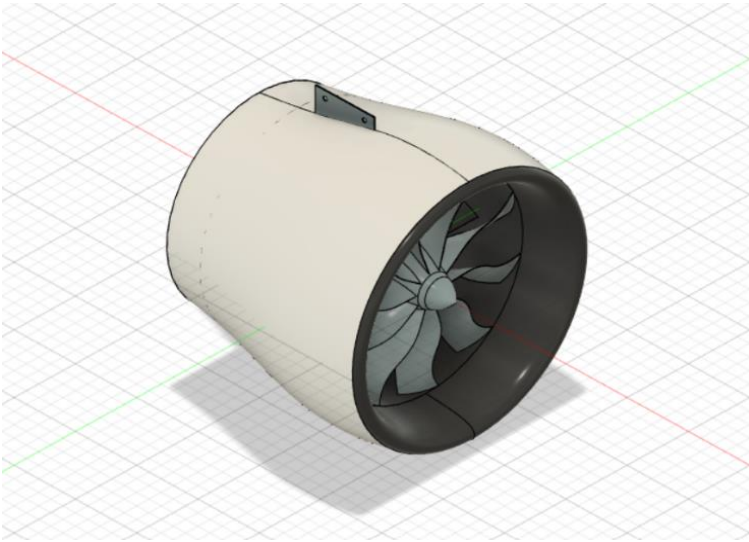


Figure 3.7

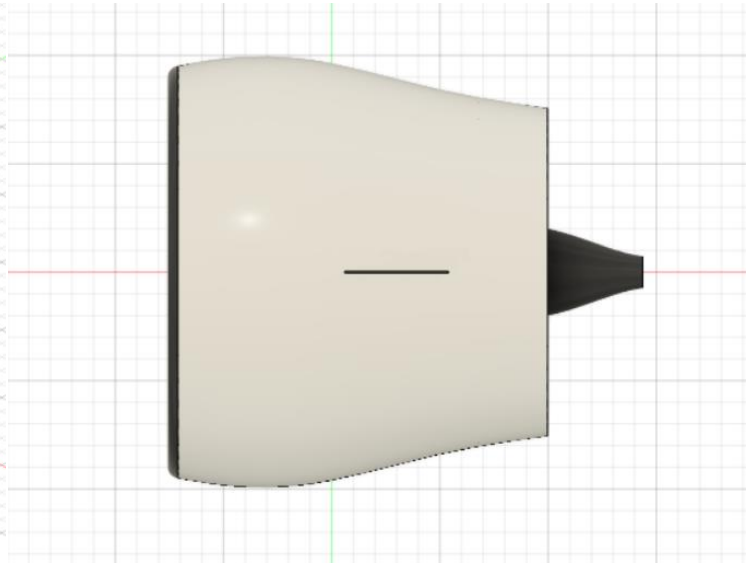


Figure 3.8

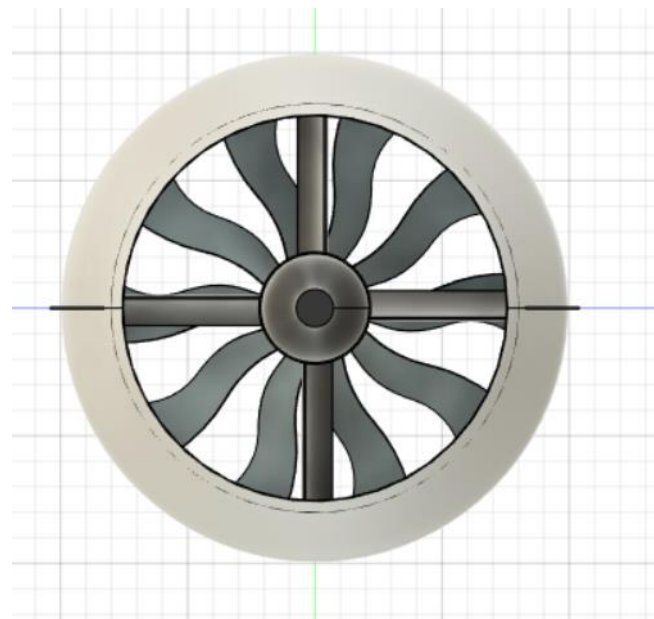


Figure 3.9



Figure 3.10

3.3 Methodology and Calculations of Engine

As there are many parameters and conditions to calculate the exact thrust generated by an Electric Ducted Fan like the shape and number of propellers, Coanda effect on the walls of EDF, Bernoulli's theorem, different properties and characteristics of air, etc. So as we are students and are learning the proper way to calculate the exact thrust, we have taken reference to an electric ducted fan from a practically tested engine VF-390 Nacelle to get exact proper data.

Practical data of VF-390

Internal diameter of the cylinder: 395 mm

Total weight of in-runner engine 12,2kg

Max. thrust range: 750N

Max. discharge speed range: 80m / s

Max. speed range: 10,000rpm

Max. input power: 45kw

Overall efficiency: 88%

Motor efficiency @ 70 - 80% of nominal power: 96%

Turns/min/Volt (Kv): 85kv

Torque 40Nm

Max Power 40kw

Below are few theoretical calculations for VF-390

Given: RPM=10000

Discharge velocity (v)= 80m/s

Internal diameter of cylinder (D)= 395mm

External diameter of hub (d)= 115.7mm

As rounds per minute =10000

So rounds per second = $10000/60 = 166.67 \dots (1)$

$D/2 = R = 0.1975\text{m}$

$d/2 = r = 0.0578\text{m}$

Area of air flow = Area of cylinder – Area of hub

$$= \pi R^2 - \pi r^2$$

$$= .1120\text{m}^2 \dots (2)$$

Volume / second = Area of air flow and Discharge velocity

$$= .1120 \text{ m}^2 \times 80 \text{ m/s} \dots \text{From(2)}$$

$$= 8.96 \text{ m}^3/\text{s} \dots\dots(3)$$

Density of Air = 1.225 kg/m^3 4

Change in mass per second = (volume/sec) x density of air

$$= 8.96 \times 1.225 \dots\dots \text{From (3\&4)}$$

$$= 10.93 \text{ kg/s} \dots\dots(5)$$

Force = mass x acceleration(by newton's second law)

$$= \text{mass} \times (\text{velocity}/\text{second})$$

$$= (\text{mass}/\text{second}) \times \text{velocity}$$

$$= 10.93 \times 80 \dots \text{From (5)}$$

$$= 874.4 \text{ N}$$

Error % in theoretical and practical data = $(874.4 - 750) \times 100 / 874.4$

$$= 14.22\%$$

Weight

Weight of paramedic and patient = 170kg

Weight of medical Equipments = 180 kg

Weight of structure = 550kg

Weight of battery = 400kg

Total Weight = 1300 kg

max thrust required = $1300 + (75\%) = 2275 \text{ kg} = 22295 \text{ N}$

Thrust generated by 1 engine = 750N

Total thrust generated by 30 engines = 22,500N

3.4 CFD Analysis of the Engine:

The below diagram is the results of the Ansys Fluent analysis which includes

- Velocity Contour
- Pressure Contour
- Velocity Contour around the fan
- Streamlined Velocity

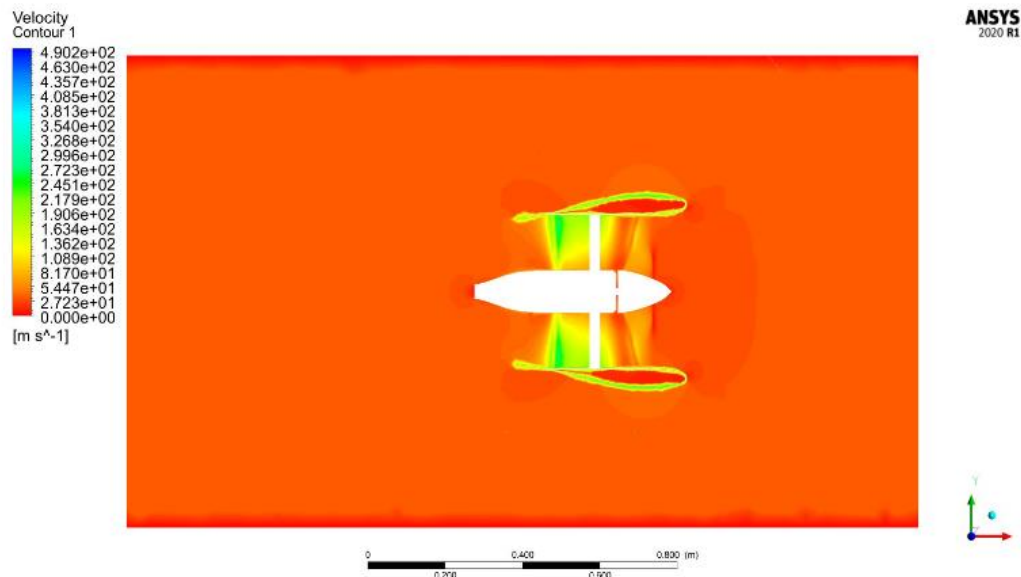


Figure 3.11 CFD of velocity Contour

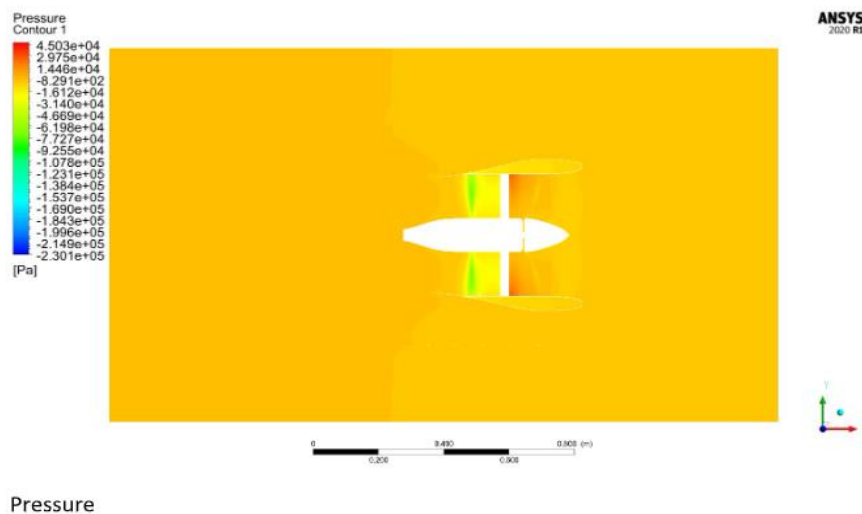
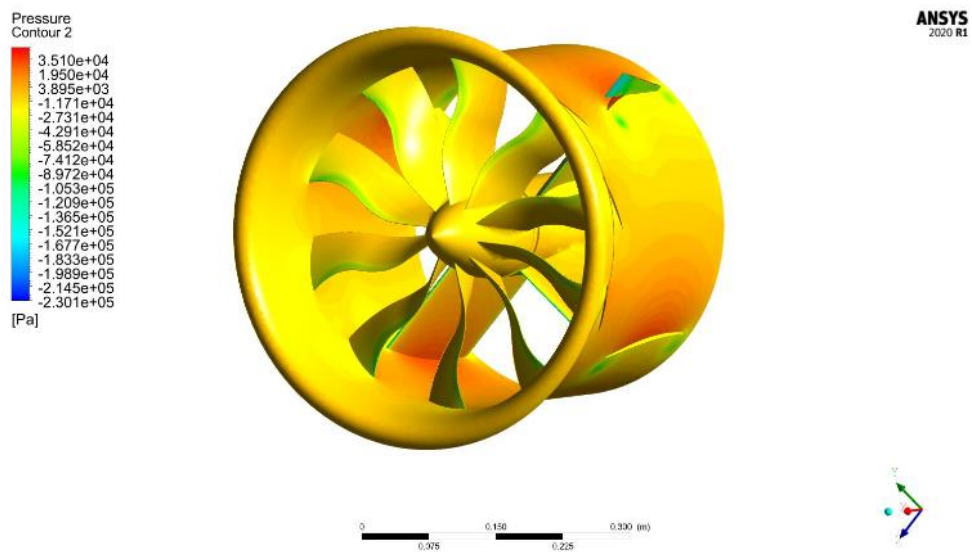
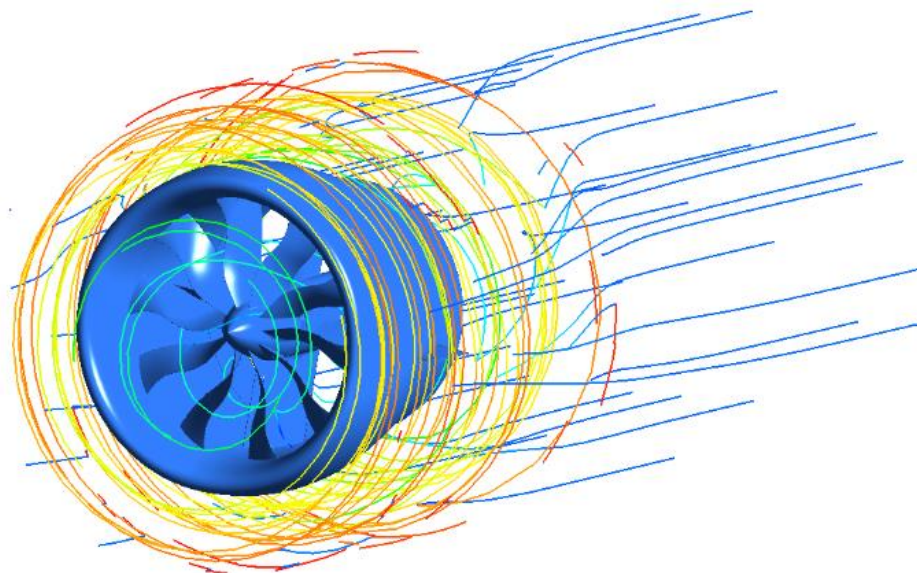


Figure 3.12 CFD of Pressure Contour



Pressure around the fan

Figure 3.13 CFD of velocity Contour around the fan



Streamlined velocity

Figure 3.14 CFD of Streamlined Velocity

The simulation of the model was done in Ansys Fluent 2020 R1 version. The K-omega model was used to simulate the case model and taking the boundary condition of the front wing, rear wing, landing gear and the fuselage to be as walls while making the rest of the enclosure as fluid medium (air, with density of 1.225 kg/m^3). The inlet velocity was taken to be 33.333 m/s (120 Km/hrs) as that is the required constraint of our vehicle during cruise.

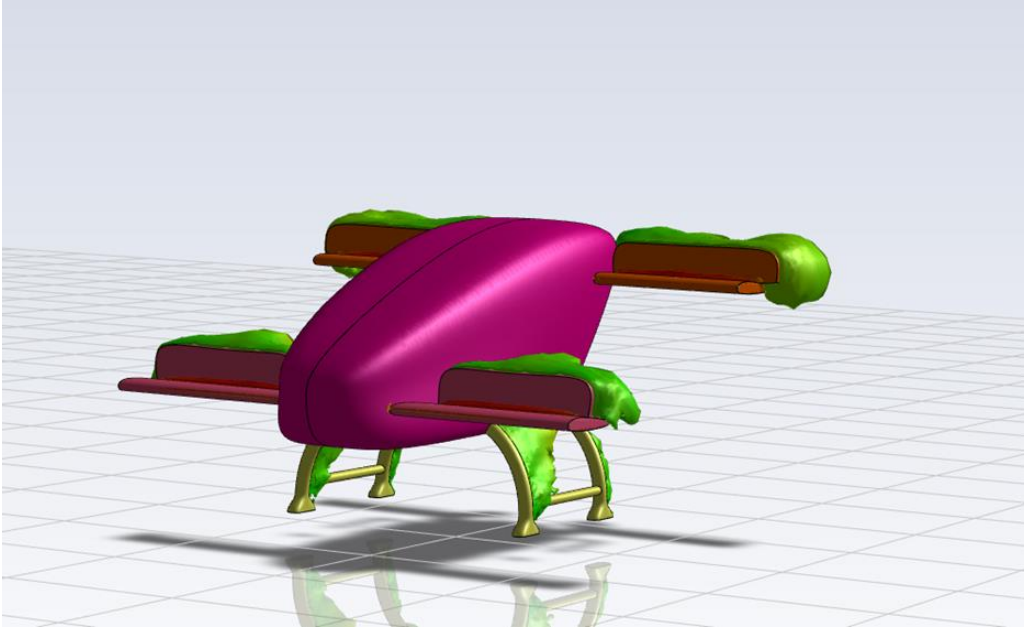


Figure 3.15 Iso-surface showing higher pressure

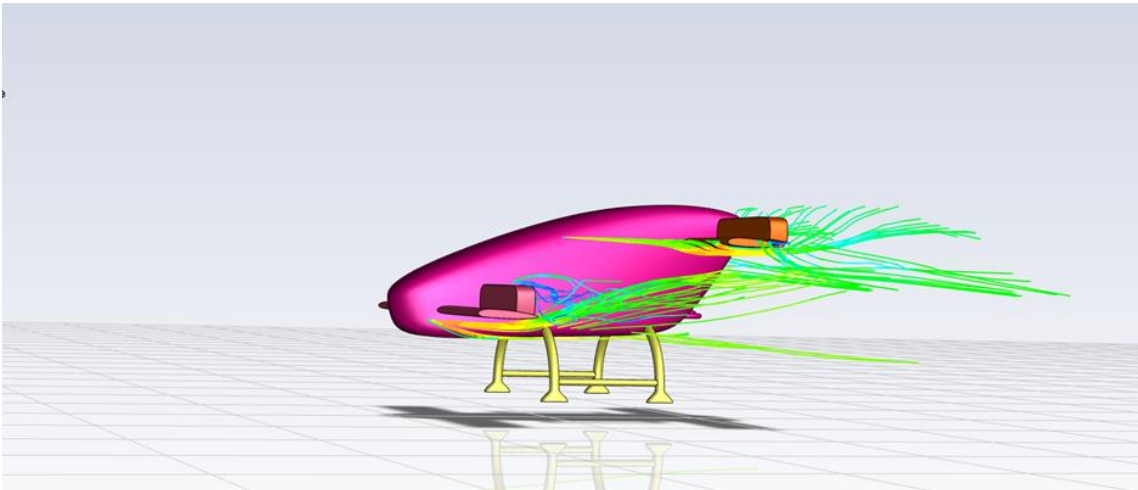


Figure 3.16 Streamlines showing velocity magnitude around the front and rear wing and the recirculation region.

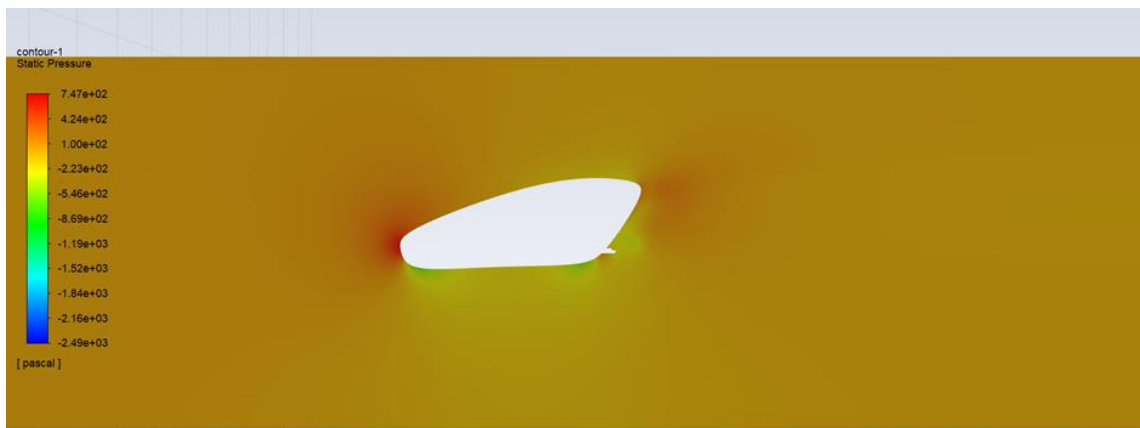


Figure 3.17 Pressure Contour showing higher pressure and lower pressure around the vehicle body

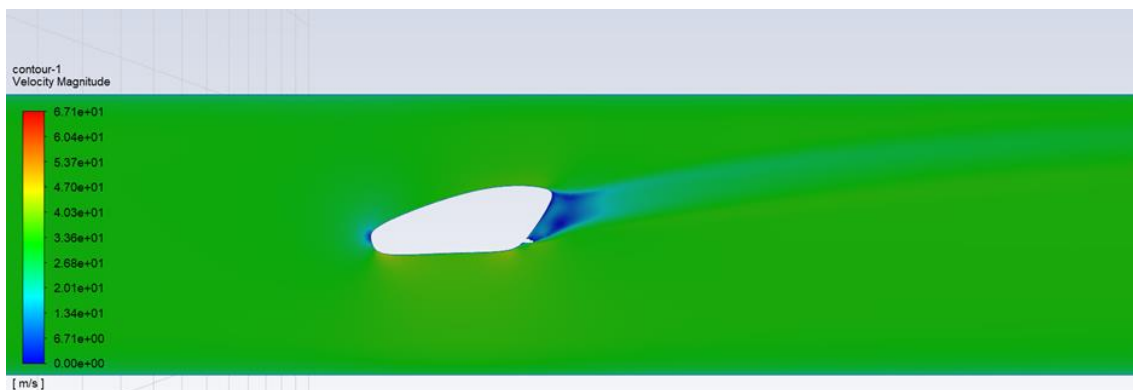


Figure 3.18 Velocity Contour showing higher pressure and lower pressure around the vehicle body

3.5 Distributed propulsion system

Instead of having large ducted fans which lead to structural challenges and increased cruise flight drag. This high disc load design leads to the freedom of having 30 smaller ducted fans and embedding them efficiently in the rear wing of our canard aircraft concept. Smaller embedded fans lead to less wetted nacelle surface area, which increases the range due to less aerodynamic drag. This concept is known as distributed electric propulsion. We can exploit this even further by controlling the aircraft using thrust vectoring without any need for standard aerodynamic control surfaces such as tails, ailerons and rudders. This essential feature reduces structural weight, aerodynamic drag and structural complexity for the whole aircraft, with positive consequences for mission performance. Since the fan is mounted behind the wing, it adds constant flow during transitional flight. The result is an efficient flow field with high lift

and excellent control during critical transition phases of flight. The greater the mass flow of the engine, the greater the lift generated. Combined with the fixed-wing structure, this high-lift effect allows us to approximate the reduction in power consumption during transition.

Engines in front wing = 14

Engines in rear wing = 16

4. Aircraft Sizing and Weight Estimations

4.1 WEIGHT ESTIMATION

The maximum takeoff weight of the aircraft is 1300 kg (about 2866.01 lb.). The weight distribution summarized as follows:

- Patient and Paramedic: 170 kg (13.1% of the total weight)
- Equipment: 180 kg (13.8% of the total weight)
- Empty Weight: 600 kg (46.2% of the total weight)
- Battery: 300 kg (23.1% of the total weight)
- Reserved Battery: 50 kg (3.8% of the total weight)
- Max thrust required = $1300 + (75\%) = 2275 \text{ kg} = 22295 \text{ N}$
- Thrust generated by 1 engine = 750N
- Total thrust generated by 30 engines = 22,500N

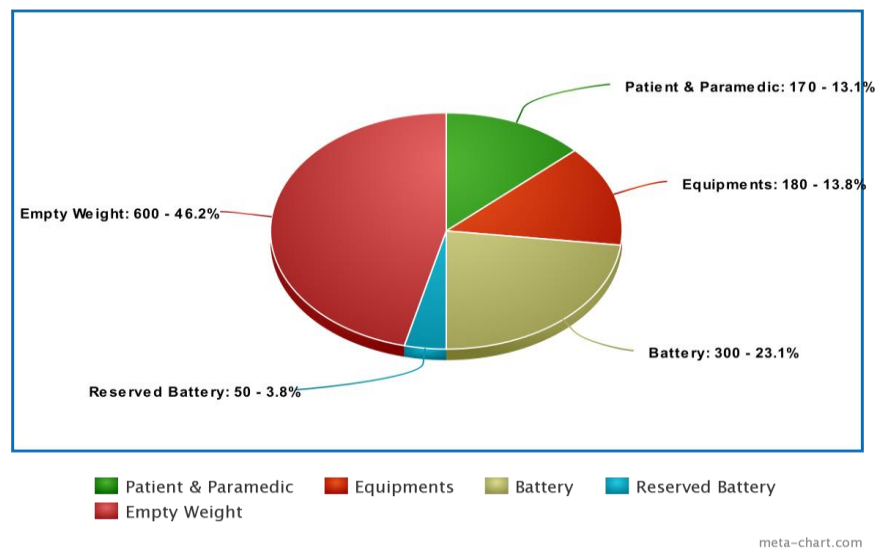


Fig:3.19- Pie Chart of the Weight Distribution of the Aircraft

The justification for this weight distribution is as follows:

- The weight of the patient and paramedic is the primary concern in the aircraft. Therefore, it is given the highest priority in terms of weight allocation.
- The equipment's weight is also important for the proper functioning of the aircraft and for providing necessary medical support to the patient.
- The empty weight of the aircraft includes the weight of the fuselage, wings, landing gear, and other structures. This weight is necessary for the aircraft to fly and maintain its stability.
- The battery weight is critical for the electric propulsion system of the aircraft. It provides the power to the electric motors, which drive the ducted fan engines.

This weight distribution is optimized for low-range, low-speed, and low-altitude EVTOL aircraft operations. Our goal is to maintain a safe and stable flight while providing necessary medical support to the patient.

4.2 Sizing of the aircraft

4.2.1 Initial sizing:

Initial sizing is a very important aspect in the design as it will affect the weight estimation and other aspects. Estimation of its design take-off gross weight W_o . Weight at the start of the design mission profile. Mission Profile specified by the user. Additional Requirements by Regulatory Bodies. Identify requirements that are likely to drive the design. First estimate of the size of the aircraft, through (W_o). And then other aspects like the wing loading, Air foil selection, drag estimation, fuselage dimensions, material selection and other aspects as well. Combining all these aspects, the aircraft can be designed perfectly, and the initial sizing is the very first step toward it.

4.2.2 Mission profile:

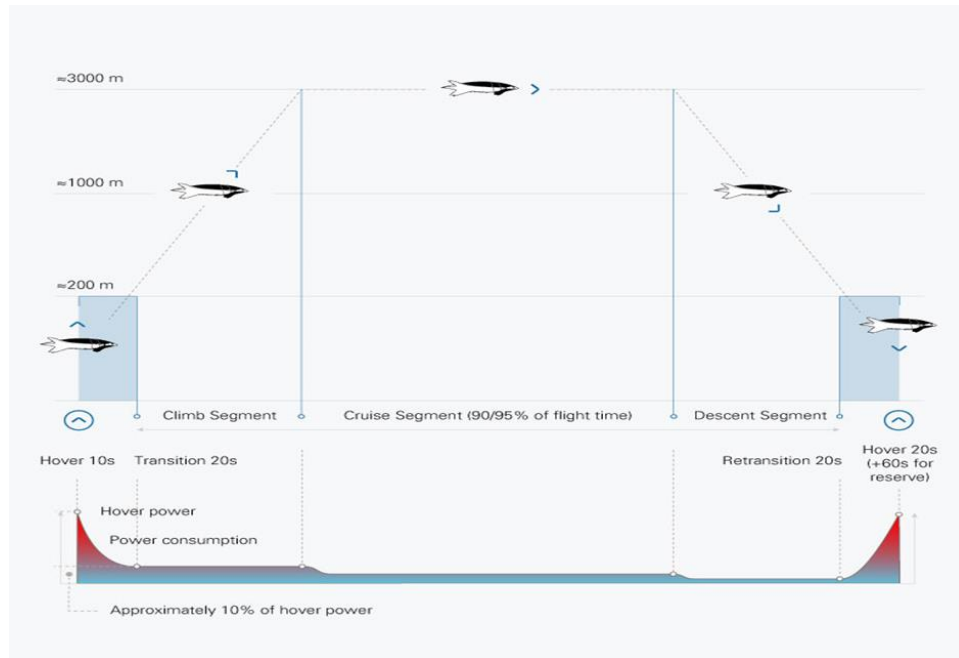


Fig- 4.1 (Mission Profile)

There are many other types of mission profile which other aircraft follows but according to our designed mission statement and requirements we choose this to be our best model of mission profile.

4.2.3 Issues of Initial Sizing:

- a. Very little known about a/c configuration.
- b. Most methods are deeply rooted in past
 - Statistical inference of parameters
 - Similar aircraft designed earlier.
- c. Most procedures empirical / semi-empirical
- d. Various methodologies / approaches
 - Loftin's method
 - Raymer's approach

From our data set we had made the graphs for the initial sizing to find out our appropriate data requirement for our aircraft

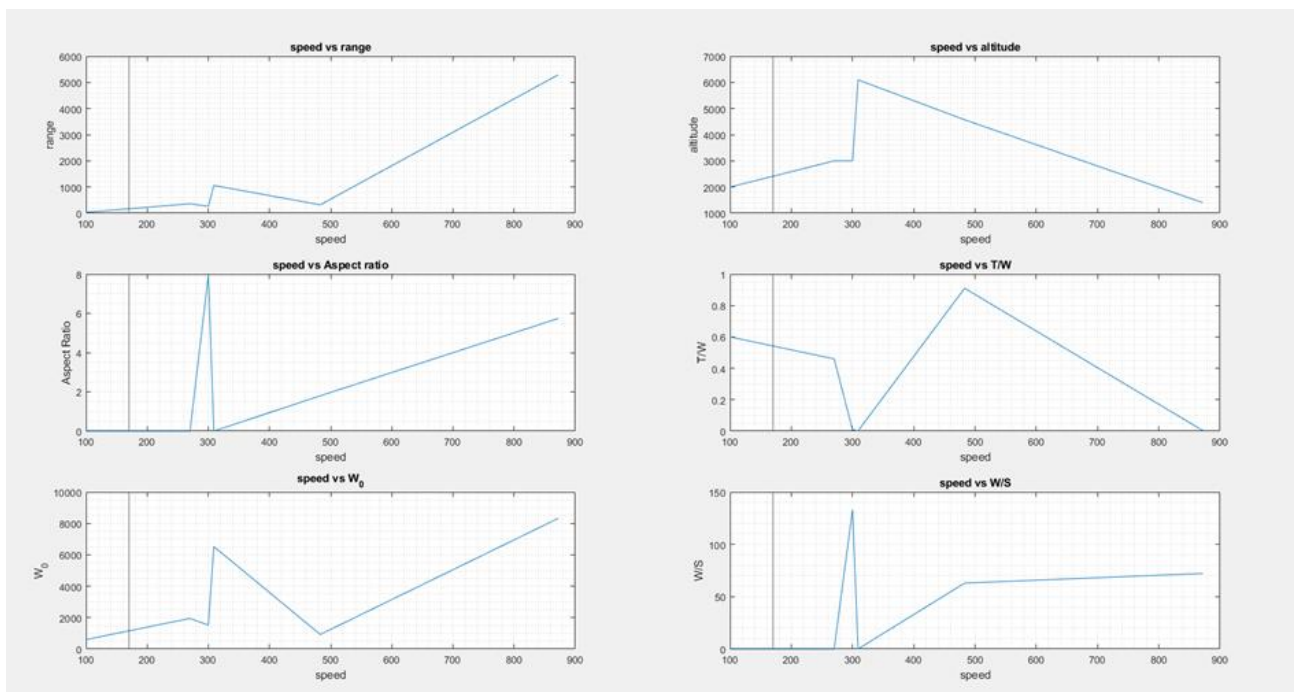


Fig- 4.2 (Initial Sizing)

Taking the velocity at cruise to be at 170 km/h. Plotting the line at 170 km/h we have got some optimized value for wing loading, range, etc. In some parts due to unavailability of data by certain conceptual aircraft. The data are assumed at that point by looking at Raymers textbook for small aircraft as our new aircraft falls in that category.

The aircraft features a tandem wing configuration with a front wing of 9 meters in wingspan, a chord of 0.7 meters, a taper ratio of 9, and an aspect ratio of 12.5. The back wing has a wingspan of 10 meters, a chord of 0.8 meters, an aspect ratio of 12.5, and a taper ratio of 0.86. The NACA 6412 airfoil is used for the design of the aircraft. These specifications will play a critical role in determining the aerodynamic performance of the aircraft, including lift and drag, stability and control, and maneuverability.

5) Fuselage Design

Design of the Fueslage

The fuselage is a crucial component in aircraft design, and its shape and structure have a significant impact on the overall performance, safety, and efficiency of an aircraft. In the case of electric vertical takeoff and landing (EVTOL) aircraft, the fuselage design becomes even more crucial as these aircraft must meet the demands of vertical flight, efficient horizontal flight, and the integration of propulsion systems and batteries.

Impact on Performance: This fuselage shape affects the aircraft's aerodynamic drag and lift, and this designed fuselage can reduce drag, increase lift, and

improve the aircraft's efficiency. In this aircraft, the fuselage is designed to accommodate the large batteries and propulsion systems while still achieving optimal aerodynamic performance in both vertical and horizontal flight.

Impact on Safety: Our fuselage design also significantly impacts the safety of an EVTOL aircraft. The fuselage must be strong enough to withstand the stress of vertical takeoff and landing, as well as the demands of high-speed horizontal flight. It is designed to protect the crew, passengers, and critical systems in a crash or other emergency.

Integration of Systems: This fuselage also accommodates the integration of propulsion systems, batteries, and other critical systems. This requires careful consideration of the placement and routing of electrical systems, fuel systems, and other components to ensure efficient and reliable operation.

5.1 Fuselage:

The fuselage, or body of the airplane, is a long hollow tube which holds all the pieces of an airplane together. The fuselage is hollow to reduce weight. As with most other parts of the airplane, the shape of the fuselage is normally determined by the mission of the aircraft. A supersonic fighter plane has a very slender, streamlined fuselage to reduce the drag associated with high-speed flight. An airliner has a wider fuselage to carry the maximum number of passengers. On an airliner, the pilots sit in a cockpit at the front of the fuselage. Passengers and cargo are carried in the rear of the fuselage and the fuel is usually stored in the wings. For a fighter plane, the cockpit is normally on top of the fuselage, weapons are carried on the wings, and the engines and fuel are placed at the rear of the fuselage.

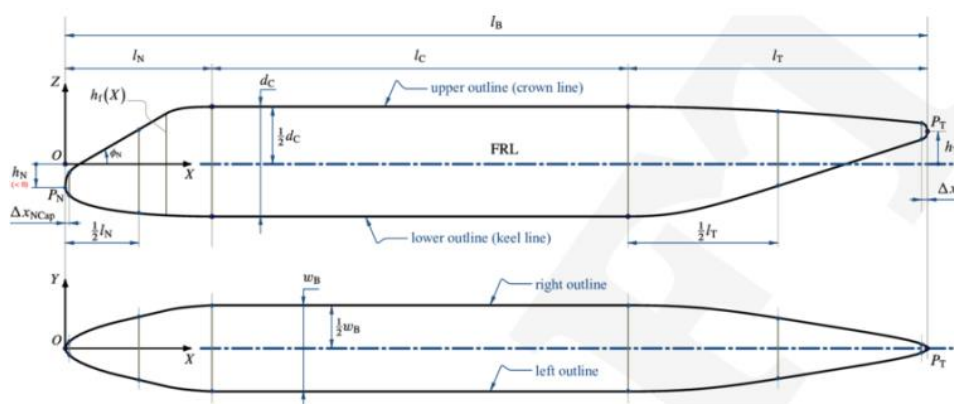


Fig- 5.1 (Fuselage)

The main function of the fuselage:

1. It shapes the aircraft and provides the necessary aerodynamics for the types of flight to be performed.
2. It serves as an assembly base for the different part of the aircraft.
3. Distributes the forces over its entire surface.
4. Protective barrier for passengers in the event of an accident.

Aviation is an industry that is always at the forefront of research and development, **so different designs for the same function are constantly emerging.**

There are therefore **several types of aircraft fuselage**, depending on force absorption (monocoque/semi-monocoque), size (wide/narrow) or method of manufacture (lattice or tubular).

Fuselages or airframes can also be classified according to their size, diameter, or internal volume.

Another type of fuselage is the lattice fuselage, which consists of a **network of steel tubes** that serve to join the frames of the aircraft. **It is one of the oldest forms of construction** and is therefore not suited to today's needs for speed and safety in flight.

5.2 Types of fuselages:

1. Truss type fuselage

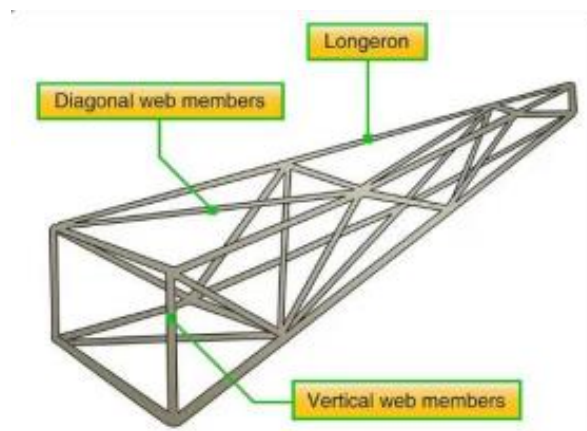


Fig- 5.2 (Truss Type Fuselage)

A truss is a structure that consists of a collection of structural elements connected at pin joints or nodes. In theory, the pin joints provide no rotational resistance and behave as hinges. In practice, this is not always the case. The benefit of a truss is that the members are predominantly axially loaded. This means they are either in compression, tension or have no force, so-called zero-force members. This makes trusses a particularly efficient structural form.

2. Monocoque type fuselage

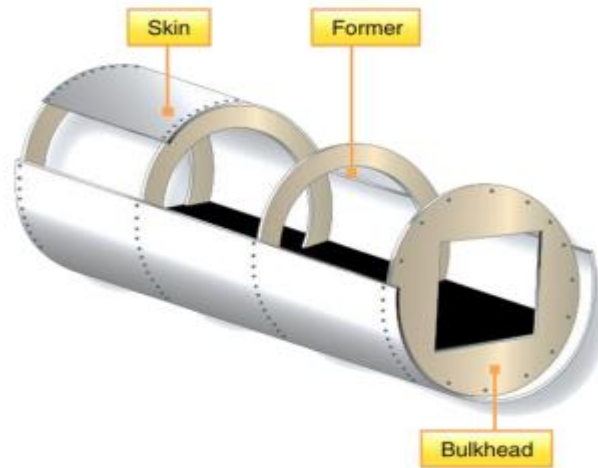


Fig- 5.3 (Monocoque type Fuselage)

In construction, monocoque is the method of using the external structure to support the load. The name literally means "single shell" and refers to fabricating an external frame into a single unit. Alternative names for this method include unibody, unitary construction, and structural skin. It is useful for significantly reducing an object's weight while maintaining its strength. The single unit design is widely used in car chassis, boat hulls, and airplane fuselages.

3. Semi- Monocoque type fuselage

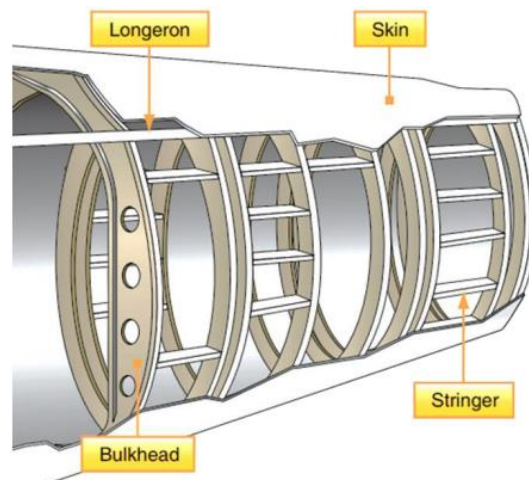


Fig- 5.4 (Semi-Monocoque type Fuselage)

The term semi-monocoque refers to a stressed shell structure that is like a true monocoque, but which derives at least some of its strength from conventional reinforcement. Semi-monocoque construction is used for, among other things, aircraft fuselages, car bodies and motorcycle frames.

Among the different types of fuselages, we opted for the Semi monocoque structure for its structural strength and has an added advantage over the monocoque structure as it the skin of the aircraft can be made light weight.

The main task of our fuselage is:

- Task is to minimize the drag and the pitching moment [make it as streamline as possible], and the volume required to accommodate the crew passengers, avionics, battery, medicine bed and avionics.
- Usage of higher taper ratio at end (streamline at rear to reduce drag) and high upsweep angle at rear (angle at which the lower portion of the rear of the fuselage curves upwards; better longitudinal stability; more neutral or slightly nose down pitching moment; desirable for stable flight).

Airplane type	Fineness ratio	L_R/D_f	θ_{UPSWEEP}
Small aircraft	4-8	3-4	2-9
Agricultural	5-8	3-4	1-7
Jet transport	6.8 – 13	2.5 – 4	15 – 19

The internal arrangement of the fuselage is:

- Depends on the modes of operational requirements.
- Cockpit design: Mostly used space in avionics and automations and sensors.
- Cabin design: Making space for the patient, paramedic, and other medical equipment's.
- Look for persons height and length, the head clearance for cabin design.

$$\text{Length of the fuselage} = \text{Length}_{\text{nose}} + \text{Length}_{\text{cabin}} + \text{Length}_{\text{rear}}$$

The table from Raymer's textbook shows the it :

Seating abreast	Nose length Ratio	Rear length Ratio	Nose Cone angle degree	Upsweep angle
≤ 3	1.7-2	2.6-3.5	10-20	12-19
4- 6	1.5 -1.8	2.5-3.75	16 – 28	
≥ 7	1.5-1.7	2.5 – 3.75	20- 30	

$$\text{Rear length ratio} = \frac{\text{Rear length}}{\text{Diameter}}$$

$$\text{Nose length ratio} = \frac{\text{Nose length}}{\text{Diameter}}$$

To find the semi- nose cone angle $\tan\left(\frac{\alpha}{2}\right) = \frac{D/2}{L_N}$ where α = nose cone angle

- Cabin length is what we have taken into consideration like the med bay length, etc.

Also, the fineness ratio is calculated from the formula $\text{FinenessRatio} = \frac{\text{Fuselage length}}{\text{Diameter}}$

From the calculations, Fineness ratio is obtained to be 5.5.

And the diameter of the fuselage is taken to be same as the average height of the human that is 1.75 m. Therefore, on calculation the fuselage length is obtained to be 9.6250 m.

The nose length ratio and rear length ratio are 2 and 2.5 as stated from the table above. Hence, we obtain the value of nose length to be 3.50 m, the rear length to be 4.3750 m, which leaves us with a cabin length of 1.750 m (enough to carry a med bed, a patient and other medical equipment's).

For finding the nose cone angle we have obtained a value of 28.0725 degrees with the modification of the formula stated above $\alpha = 2 * \tan^{-1}\left(\frac{D}{L_N}\right)$, where α is the nose cone angle.

Specific Fuselage Design of our Air Ambulance:

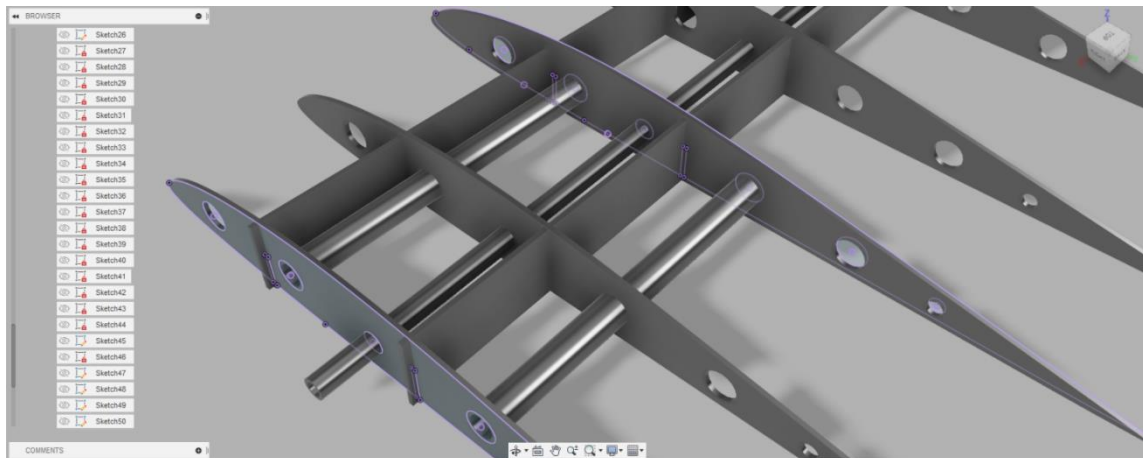


Figure : 5.4

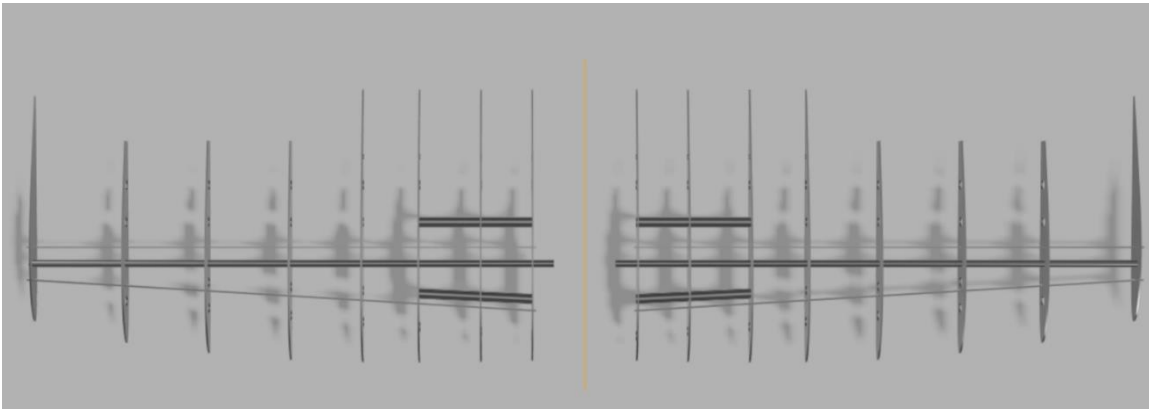


Figure : 5.4

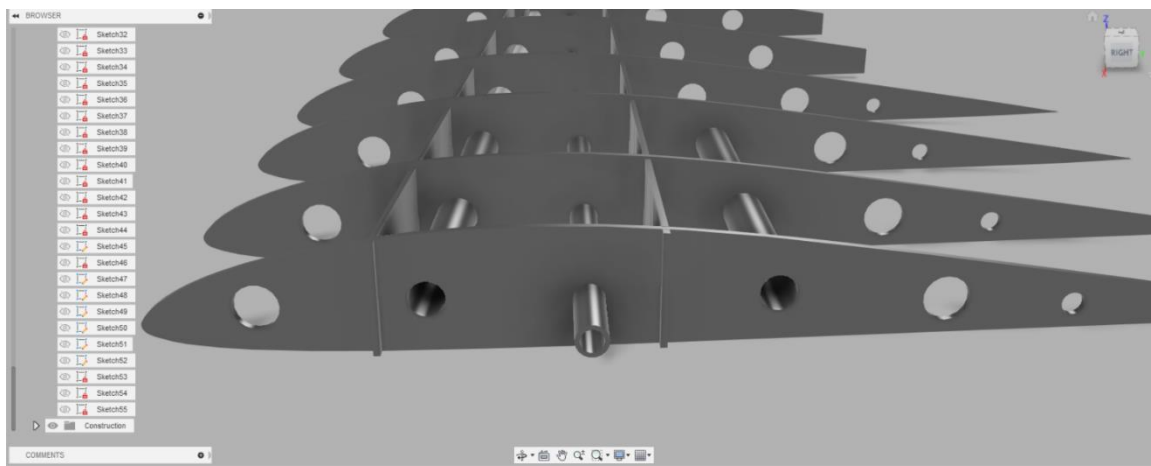


Figure : 5.5

6.WING CONFIGURATION OF THE AIRCRAFT

The wing configuration of an aircraft plays a critical role in determining its aerodynamic performance, stability, and maneuverability. The right-wing configuration can significantly impact the aircraft's ability to fly efficiently, safely, and effectively.

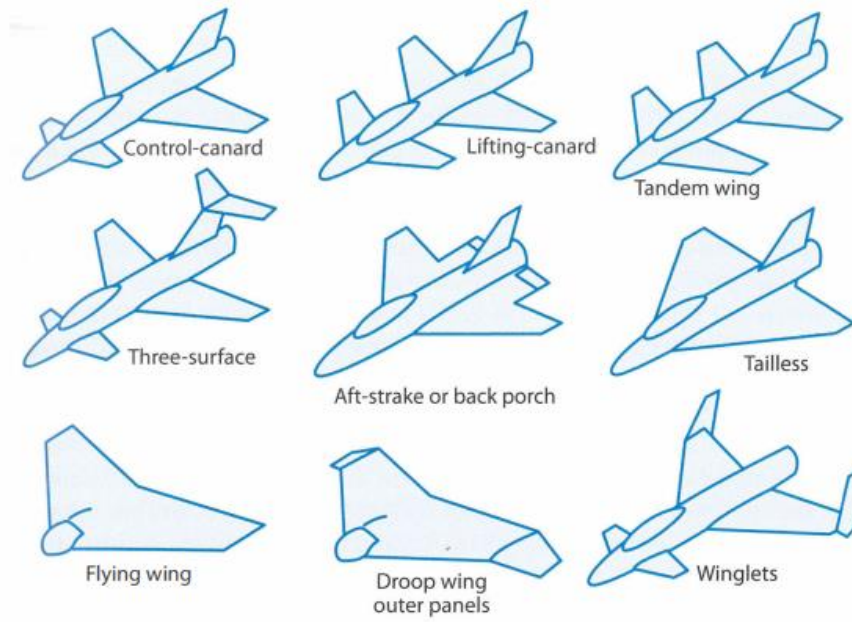


Figure: 5.6- Different Wing Configurations

The flight envelope of an aircraft is defined by its maximum and minimum flight parameters, such as speed, altitude. The flight envelope of a canard aircraft is limited by its high drag, which reduces its maximum speed. The flight envelope of a conventional wing aircraft is limited by its stability, which affects its maneuverability. The flight envelope of a tandem wing aircraft is limited by its lift capacity, which affects its maximum altitude.

After doing several analyses we found that the two best wing configurations of the aircraft is Canard Wing and Tandem Wing Arrangement.

In a tandem wing aircraft, two wings are placed one in front of the other, each with its own set of control surfaces. The tandem wing configuration provides improved stability and control compared to the conventional wing configuration.

In a canard aircraft, the wing is placed ahead of the main wing, as opposed to the conventional wing configuration where the wing is placed behind the fuselage. This configuration allows for a higher degree of control authority and improved maneuverability, but also results in increased drag and a reduced wing loading.

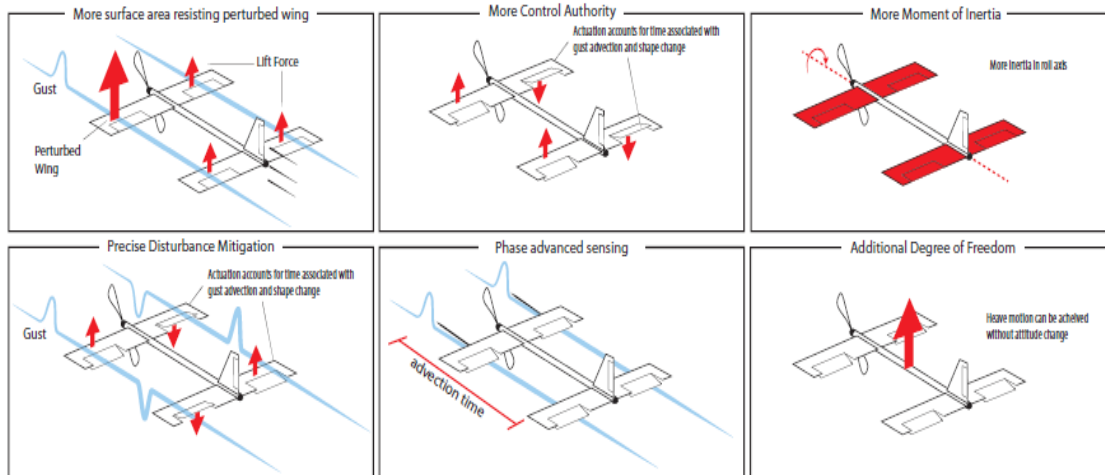


Figure: 5.7- Tandem wing aircraft hypothesized advantages with respect to counteracting turbulence that helps our aircraft cruise at different situations.

6.1 GEOMETRY OF THE WING

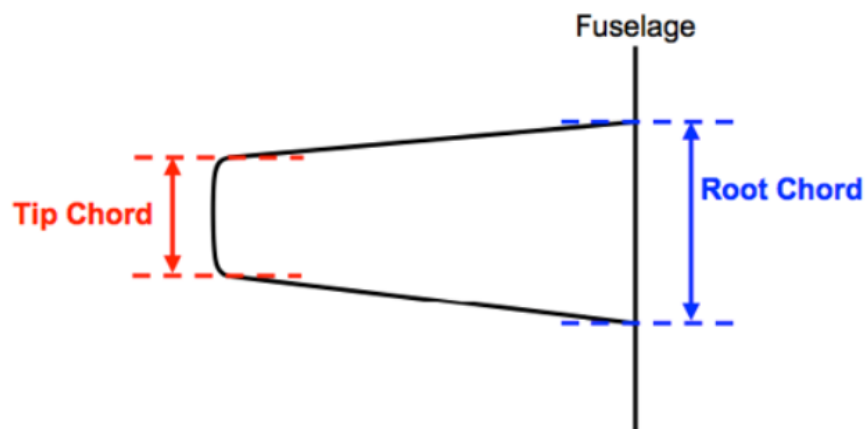


Figure: 5.8- Chord of the Wing

Front Wing

- Wingspan: 9.04 m
- Aspect Ratio: 12.5
- Taper Ratio: 9
- Root Chord: 0.7m
- Tip Chord: 6.3m

Rear Wing

- Wingspan: 10m
- Aspect Ratio: 12.5

- Taper Ratio: 0.86
- Root Chord: 0.8m
- Tip Chord: 0.68m

Choosing the Airfoil

The NACA 6412 airfoil has a symmetrical profile with a thickness of 12% of its chord length. It is known for its superior performance in generating lift, which is essential for vertical take-off and landing (VTOL) aircrafts. The symmetrical profile of the NACA 6412 provides it with a stable and predictable lift coefficient, making it a good option for VTOL aircrafts, especially those with low payloads, which need to lift off the ground with minimum lift.

Another advantage of the NACA 6412 airfoil is that it provides a good balance between lift and drag, making it suitable for low altitude flights. The airfoil has a lower induced drag coefficient compared to other airfoils, which is essential for low altitude flights where the aircraft is operating in denser air, which results in higher drag.

The NACA 6412 airfoil has a high lift coefficient, which is important for VTOL aircrafts. The high lift coefficient ensures that the aircraft can generate enough lift to take off vertically, and the symmetrical profile ensures that the lift coefficient remains predictable, even at high angles of attack.

NACA 6412 airfoil is considered as a good option for VTOL aircrafts due to its symmetrical profile, good lift to drag ratio, and high lift coefficient. These characteristics make it a suitable option for low payload aircrafts, operating at low altitudes, and needing high lift.

α°	h/c	C_l	C_d	C_l/C_d
0	0.2	0.67112	0.012566	53.41
0	0.5	0.65475	0.012614	51.66
0	1.0	0.65437	0.013279	49.28
4	0.2	1.1841	0.015862	74.65
4	0.5	1.0658	0.019398	54.94
4	1.0	1.0397	0.020825	49.93
8	0.2	1.5217	0.031304	48.61
8	0.5	1.3245	0.034725	38.14
8	1.0	1.3183	0.034992	37.67

Table: 4.1- Showing C_l , C_d values of NACA 6412 airfoil at various angles of attack (α) and h/c ratio

- At an angle of attack (AOA) of 0 degrees, the NACA 6412 airfoil is said to be at its zero-lift angle, meaning that it generates no lift force. The lift coefficient (C_L) is equal to 0.0 and the drag coefficient (C_D) is at its minimum value.
- At an angle of attack of 4 degrees, the NACA 6412 airfoil starts to generate lift and the lift coefficient increases. The drag coefficient also increases as the airfoil starts to generate more turbulence and pressure drag. The lift-to-drag ratio (L/D) of the airfoil is likely to be at its highest value at this angle of attack.
- At an angle of attack of 10 degrees, the lift coefficient continues to increase, but the drag coefficient starts to increase at a faster rate. The airfoil may start to stall at this angle of attack, meaning that the lift force decreases suddenly and the drag force increases.

6.2 DEGREE OF FREEDOM

The number of degrees of freedom for this EVTOL aircraft are: pitch (nose up/down), roll (wing up/down), yaw (nose left/right), heave (vertical movement), surge (forward/backward), and sway (left/right). This exact number of degrees of freedom may also be influenced by the presence of additional systems such as landing gear or control surfaces.

6.3 TRADE OFF ANALYSIS WITH BACKUP REASONING

The purpose of this trade-off analysis is to evaluate and compare different design options for an aircraft project with the goal of selecting the optimal design that best meets the desired performance, cost, weight, and safety requirements. This trade-off analysis will consider various trade-offs between conflicting design requirements, such as weight and performance, and will provide a comprehensive evaluation of the trade-offs involved in each design option. The results of this analysis will be used to support the selection of the final design for the aircraft.

Project Overview: The aircraft design project aims to develop an efficient, safe, and cost-effective aircraft for a specific application as air ambulance. The project will consider various design options, such as the wing configuration, the propulsion system, and the airframe structure, and will evaluate the trade-offs between the different options based on their impact on the aircraft's performance, cost, weight, and safety.

Trade-Off Analysis: The trade-off analysis will provide a comprehensive evaluation of the trade-offs involved in each design option, and will help to identify the optimal design that best meets the desired requirements. The analysis will consider factors such as the aircraft's takeoff and landing performance, cruise speed, range, payload capacity, weight, and operating costs,

and will provide a detailed comparison of the trade-offs between the different design options.

1. Several Electric engines v/s Few Large Engine

Advantages of Several Small Electric Engines:

- Improved redundancy and reliability, as the failure of one engine will have a reduced impact on the overall performance of the aircraft
- Improved power density, as the small size of the engines reduces the weight and size of the power systems, potentially increasing the payload and range of the EVTOL
- Lower initial cost, as the smaller engines may be less expensive to manufacture and install

Disadvantages of Several Small Electric Engines:

- Increased complexity and maintenance costs, as there are more engines and systems to manage and maintain
- Increased weight and drag, as the installation of several engines increases the size and weight of the aircraft, potentially reducing the performance and range of the EVTOL

Advantages of Few Large Electric Engines:

- Simplified maintenance, as there are fewer engines to manage and maintain
- Improved efficiency, as the large size of the engines increases the power density and reduces the weight of the power systems, potentially increasing the range and payload of the EVTOL
- Improved performance, as the large size of the engines increases the power output, potentially increasing the speed and maneuverability of the EVTOL

Disadvantages of Few Large Electric Engines:

- Reduced redundancy and reliability, as the failure of one engine will have a significant impact on the overall performance of the aircraft
- Increased initial cost, as the large engines may be more expensive to manufacture and install
- Increased weight and drag, as the large size of the engines increases the size and weight of the aircraft, potentially reducing the performance and range of the EVTOL

2. NACA 6412 v/s Other Airfoils

Advantages of the NACA 6412 Airfoil:

- Good lift performance at low to medium Reynolds numbers
- Good stall behavior, with a well-defined stall point
- Good balance between lift and drag, making it a good choice for general-purpose aircraft

Disadvantages of the NACA 6412 Airfoil:

- Limited high-speed performance compared to other airfoils optimized for high-speed flight
- May not be the best choice for specialized aircraft, such as gliders or high-speed jets

Advantages of Low-Reynolds Number Airfoils:

- Good lift performance at low Reynolds numbers, making them well-suited for unmanned aerial vehicles (UAVs) and small aircraft
- Lower drag compared to high-Reynolds number airfoils, leading to improved fuel efficiency

Disadvantages of Low-Reynolds Number Airfoils:

- Limited high-speed performance compared to high-Reynolds number airfoils
- May not be the best choice for larger aircraft with high lift requirements

Advantages of High-Speed Airfoils:

- Good lift performance at high Reynolds numbers, making them well-suited for high-speed aircraft
- Improved high-speed performance compared to general-purpose airfoils

Disadvantages of High-Speed Airfoils:

- Increased drag compared to low-Reynolds number airfoils, leading to reduced fuel efficiency
- May not be the best choice for low-speed aircraft or UAVs

3. High Empty Weight v/s Low Empty Weight

Advantages of High Empty Weight:

- Increased structural strength and stability, as a heavier aircraft is less likely to be impacted by turbulence and wind gusts, potentially improving the safety and reliability of the EVTOL
- Improved fuel efficiency, as a heavier aircraft can fly more efficiently and with less energy, potentially increasing the range and reducing the operating costs of the EVTOL
- Increased payload capacity, as a heavier aircraft can carry more weight and equipment, potentially increasing the operational versatility and value of the EVTOL

Disadvantages of High Empty Weight:

- Reduced performance, as a heavier aircraft has a lower power-to-weight ratio and requires more energy to fly, potentially reducing the speed and maneuverability of the EVTOL
- Increased operating costs, as a heavier aircraft requires more energy to fly and more maintenance to support, potentially increasing the operating costs and reducing the profitability of the EVTOL
- Reduced range, as a heavier aircraft requires more energy to fly and has a lower endurance, potentially reducing the operational versatility and value of the EVTOL

Advantages of Low Empty Weight:

- Improved performance, as a lighter aircraft has a higher power-to-weight ratio and requires less energy to fly, potentially increasing the speed and maneuverability of the EVTOL
- Reduced operating costs, as a lighter aircraft requires less energy to fly and less maintenance to support, potentially reducing the operating costs and increasing the profitability of the EVTOL
- Increased range, as a lighter aircraft requires less energy to fly and has a higher endurance, potentially increasing the operational versatility and value of the EVTOL

Disadvantages of Low Empty Weight:

- Reduced structural strength and stability, as a lighter aircraft is more likely to be impacted by turbulence and wind gusts, potentially reducing the safety and reliability of the EVTOL
- Reduced fuel efficiency, as a lighter aircraft flies less efficiently and with more energy, potentially reducing the range and increasing the operating costs of the EVTOL
- Reduced payload capacity, as a lighter aircraft can carry less weight and equipment, potentially reducing the operational versatility and value of the EVTOL

4. Tandem Wing v/s Other Wings

Advantages of the Tandem Wing Configuration:

- Increased lift-to-drag ratio, leading to improved fuel efficiency
- Good stability and control characteristics at high angles of attack
- Reduced stall speed, allowing for operation in hot and high environments
- Improved visibility for the pilot compared to conventional wings

Disadvantages of the Tandem Wing Configuration:

- Increased structural complexity and weight
- Potential for interference between the front and rear wings
- Limited design experience, as this configuration is not commonly used in commercial aircraft

- Potential for reduced stability compared to conventional wings

Advantages of Conventional Wing Configuration:

- Proven design with a long history of use in commercial and military aircraft
- Easy to manufacture and maintain
- Improved stability compared to the tandem wing configuration
- Well-understood aerodynamics and performance characteristics

Disadvantages of Conventional Wing Configuration:

- Lower lift-to-drag ratio compared to the tandem wing configuration, leading to reduced fuel efficiency
- Potential for higher stall speed in hot and high environments

Advantages of Delta Wing Configuration:

- High lift-to-drag ratio, leading to improved fuel efficiency
- Good stability and control characteristics at high angles of attack
- Good short-field performance
- Potential for reduced weight compared to the tandem wing configuration

Disadvantages of Delta Wing Configuration:

- Complex aerodynamics, leading to reduced stability at low speeds
- Potential for increased weight compared to conventional wings
- Limited design experience, as this configuration is not commonly used in commercial aircraft

7) Landing Gear:

The main purpose of the Landing Gear is to help the aircraft to have a safe and stable landing during touch down. But in the case of VTOL (Vertical Take Off and Landing), the basic concept is same as of the conventional Landing Gears, but the only difference is, VTOL don't require any runway or any other strip for runup, instead they just land on a particular spot designated to them.

For landing they use special kind of Landing Gears that are partially retractable (Only extended at the ends for soft landing) and has an elongated base plate on each arm for stability and increased surface area. The Extended retracts arm behave exactly like the normal conventional retracts but in the conventional landing gear there are 2-axis motions i.e., vertical as well as horizontal (vertical for soft landing with tyres and horizontal for retracting the landing gear out from fuselage), but in this case the landing gear works only in one axis i.e., only vertically. The extended arm works as the retracts as well as the suspension for soft landing. The basic idea is, when the VTOL is taking off/hovering the extended arm gets retracted when it lifts off and when it is landing, the retracts gets out and

helps in soft landing (as suspensions). These landing gears works on the principle of either by pneumatics system (Air based) or by hydraulic system (Oil based).

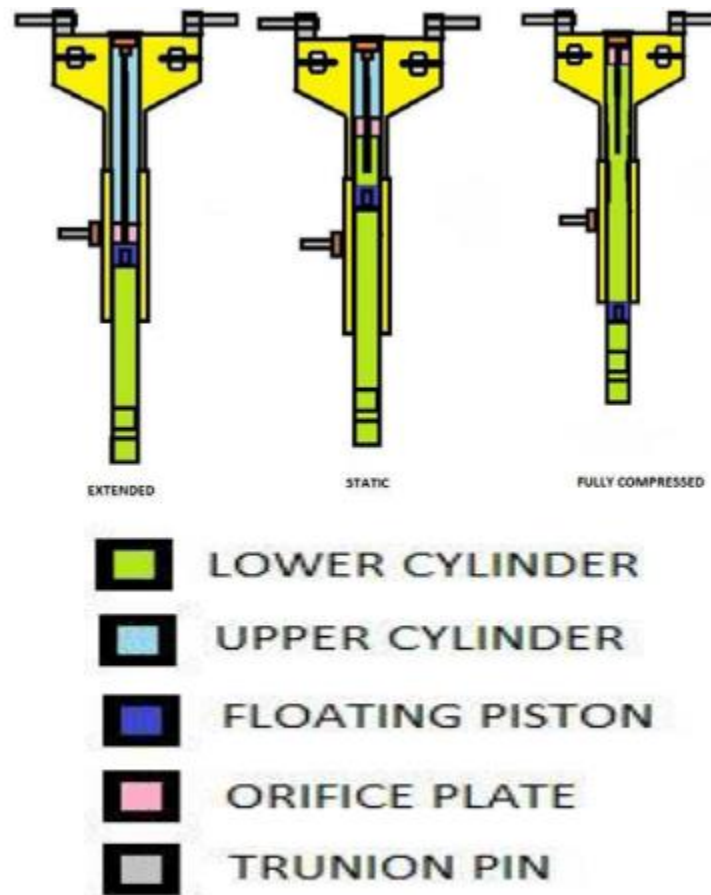


Fig- 7.1 (Landing Gear)

8 .Energy Distribution and Battery System

8.1 Battery technology and its energy consumption :

When we think of the EV based or hybrid aircraft we always think about the electricity as the power of source and for hybrid combining both fuel and electricity but the electricity is mostly carried out from the batteries which is heavy and takes lots of time to recharge it .To avoid the barriers and make more efficient we are adapting our own hydrogen fuel cell based battery which will power the our ducted fans more efficiently and less power consuming comparing to the traditional lithium batteries . Hydrogen fuel cells combine hydrogen and oxygen to produce electricity.

Energy cells give advantages over traditional combustion- grounded technologies, including lesser edge and lower emigrations. Since hydrogen

energy cells only emit water, there are no carbon dioxide emissions or other adulterants released into the atmosphere. Energy cells are also quiet during operation as they've smaller moving corridor than combustion technologies.

As numerous companies search for an emission-free solution to internal combustion machines in their forklifts, some are turning to hydrogen energy cell technology. Hydrogen energy cells don't produce any CO₂ emissions during operation, indeed if their product isn't inescapably carbon-free. This gives them an advantage over combustion machine vehicles, which can emit small quantities of toxic carbon monoxide and bear well-voiced apartments for inner use. The refueling process for hydrogen energy cells is easy and takes about three twinkles, compared to the longer and more complex process of changing out a large, heavy supereminent acid battery when it needs to be charged. In comparison to refueling a propane-powered forklift, the hydrogen energy cell refueling process has an advantage. Because the energy viscosity of hydrogen energy cells is veritably high, hydrogen-powered equipment can operate for longer times and with lower frequent refueling than with a propane cylinder. Hydrogen energy cells and lithium-ion battery energy consistency are veritably analogous, making them great choices for busy operations who need an outfit to last a full shift.

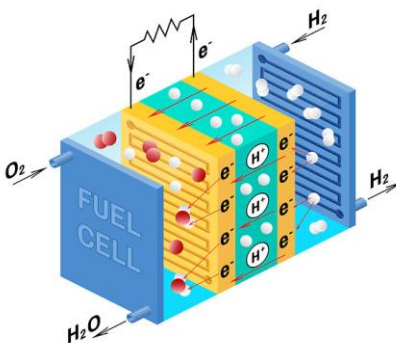


Figure 8.1 Hydrogen Fuel Cell

To find out the range that an aircraft is able to travel and cover the range we are using this range equation by how much the energy is consumed.

8.2 Schematic Layout of battery system :

The below diagram indicates schematics of the production of electricity from the hydrogen fuel cell and how it is transferring the current to the motor . The fig

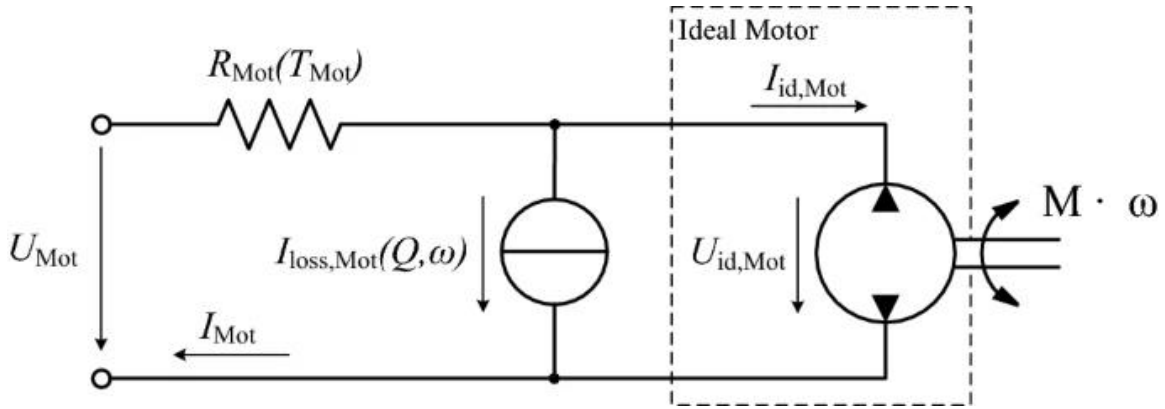


Figure 8.2

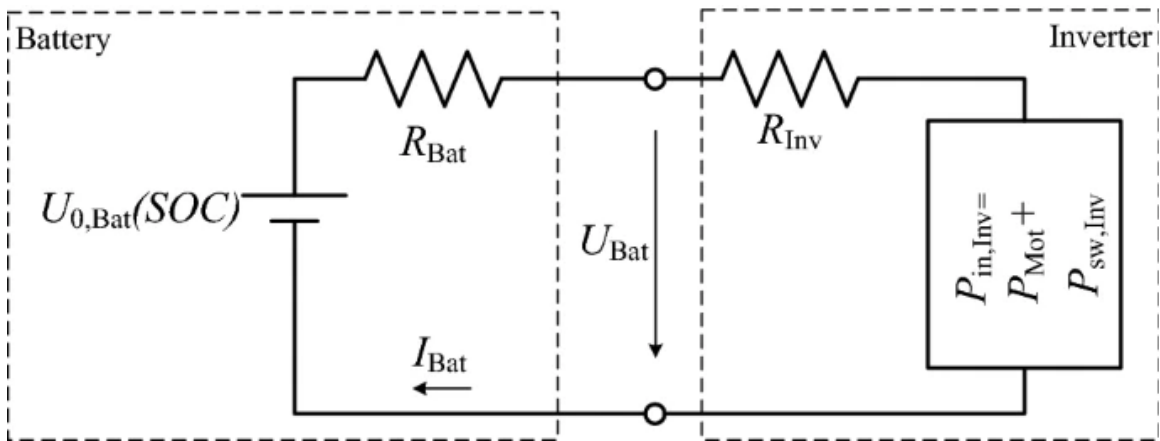


Figure 8.3

5.3 Formulation and Calculations with Graphs :

The range equation for battery electric aircraft is:

$$R = 3.6 \cdot e_b \cdot w_b \cdot \frac{1}{g} \cdot \frac{L}{D} \cdot \eta \cdot f$$

Here

R is the range in kilometres;

e_b is the specific energy of the battery in Wh/kg;

w_b is the fraction of the total mass of the aircraft occupied by the battery system;

$g = 9.81 \text{ m/s}^2$ is the acceleration due to gravity;

$\frac{L}{D}$ is the lift/drag ratio;

η is the efficiency of the battery-motor-propeller system; and

f is the available fraction of the battery capacity.

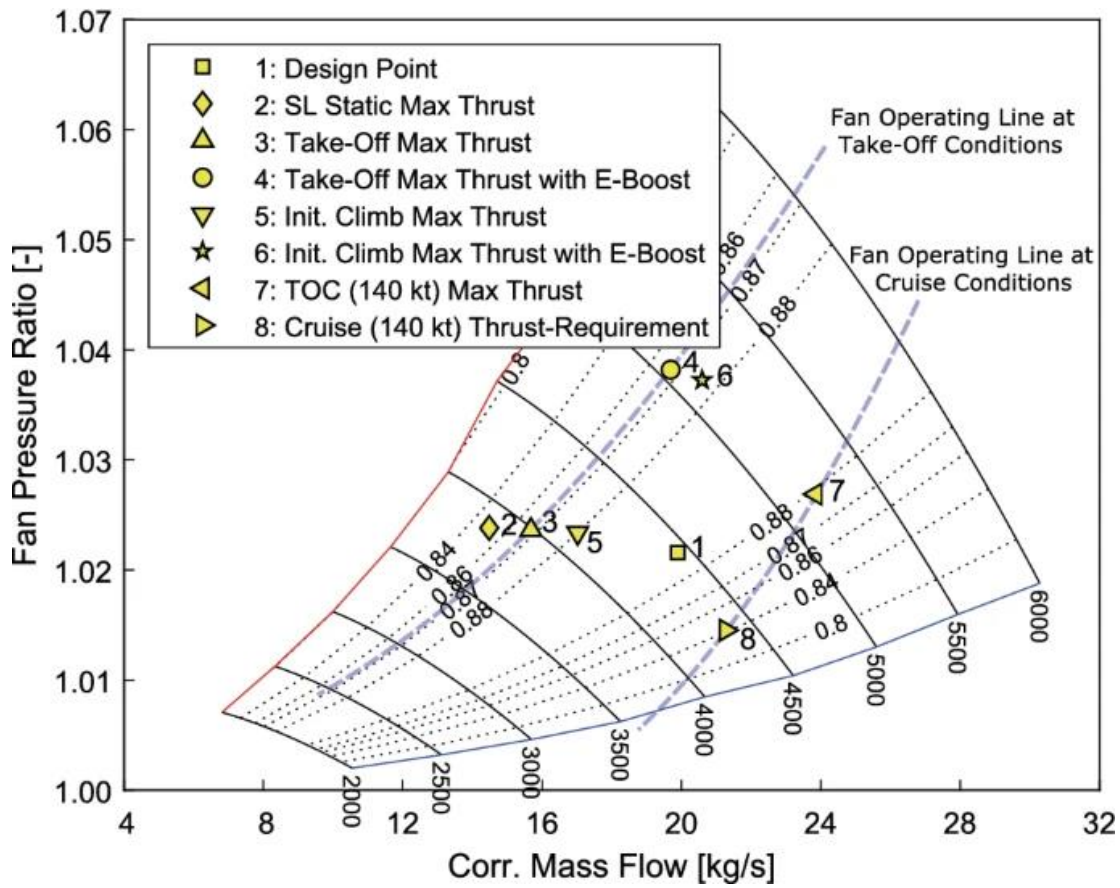


Figure 8.4

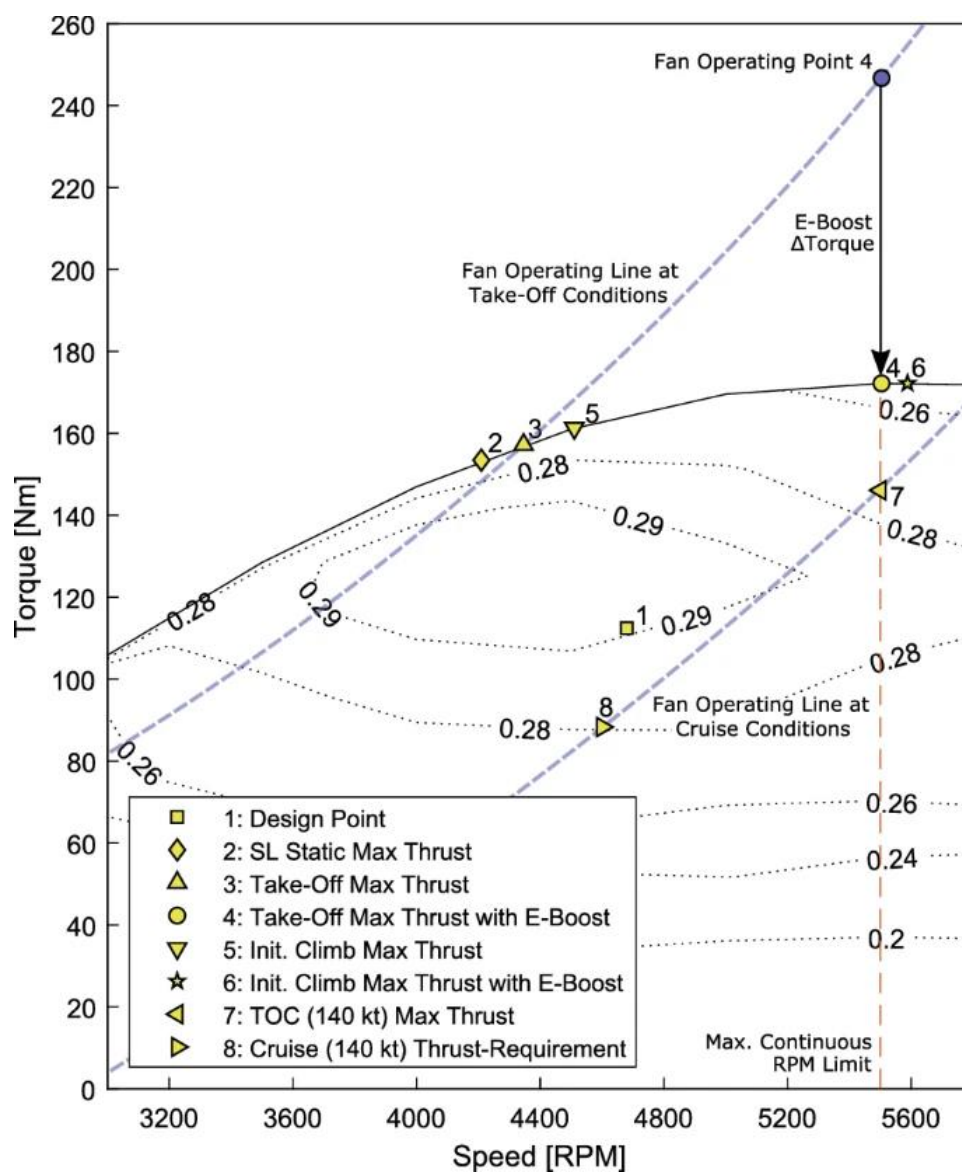


Figure 8.5

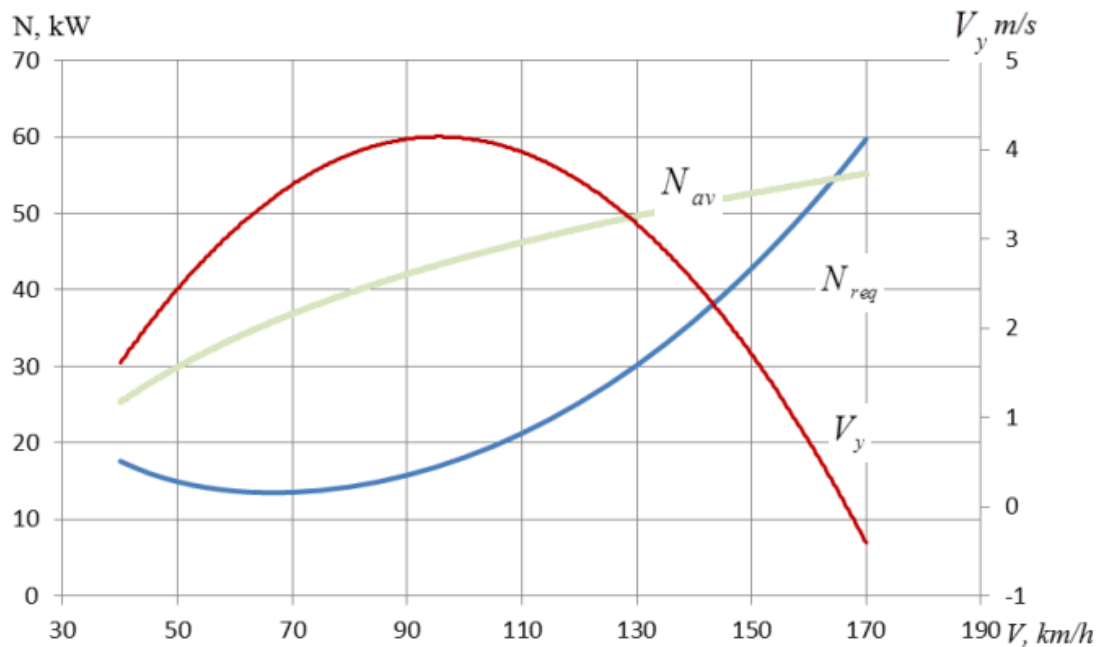


Figure 8.6 Matlab Simulation

9. Stability and Control

The aircraft's stability is the primary determinant of its ability to return to its main route. Aerial vehicles can maintain uniform flight conditions, recover from disturbances, and minimize pilot workload through the use of artificial intelligence and static stability.

Static stability is the ability of a body or aircraft to return to its initial position after it is disturbed. There are three types of static stability:

1. Positive static stability :

In positive static stability the aircraft or the body return to its initial position after small or larger disturbance.

2. Neutral static stability :

In neutral stability the aircraft or the body tends to stay at its new altitude after it is disturbed.

3. Negative static stability:

In negative stability the aircraft or the body tends to move away from the original position after it is disturbed.



Figure 9.1 Flight path

An aircraft's dynamic stability depends mainly on its static stability over time. For example the fighter aircrafts have very less stability in comparison to the commercial or transport aircrafts. There are three types of dynamic stability

1. Positive dynamic stability:

Positive dynamic stability is when the oscillations are tried to dampen out. Each and every time the oscillation factor reduces to return to its original altitude.

2. Neutral dynamic stability:

Neutral dynamic stability is the one in which if an aircraft is disturbed, the oscillations never dampens out. Hence there is no correction.

3. Negative dynamic stability:

Negative dynamic stability is the one in which if an aircraft is disturbed, the oscillation instead of decreasing, tends to increase and the aircraft becomes uncontrollable.

Talking about the combined effect of degrees of freedom and the stability factors, static stability is mostly dependent on the center of gravity of the aircraft and the neutral point. The neutral point is that point where the pitching moment coefficient is independent of the angle of attack.

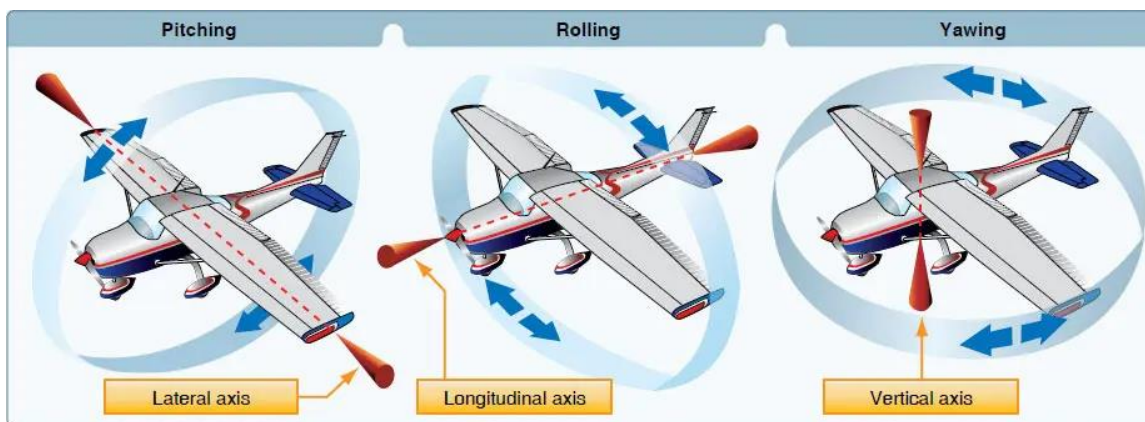


Figure 9.2 showing stability in conventional aircraft and its degrees of freedom.

In the case of the E-VTOL, the engines are mounted mostly on the flaps of the wings and the pitching moment, rolling and turning moment are mostly performed by the thrust vectoring. The VTOL lacks a vertical stabilizer, so control is mostly performed by the engines. During the take-off and landing phase of the aircraft, the aircraft needs a vertical lift factor without a runway. So, this factor is solved by the engines on the flaps. For take-off and hover, all the flaps (both side of the rear wing and both sides on the front wing) are deflected at right angles to the wings. This makes the engine produce thrust in the upward direction which produces the net forces to lift the aircraft.

Though in conventional or other aircrafts with single engines mounted at the front nose, the cg of the aircraft is mostly situated after the main wings. This makes the main wings less stabilizing. When there is a gust of wind approaching the aircraft creates an angle of attack on the wing, the aerodynamic center located at the quarter chord point of the aircraft causes a moment about the CG of the aircraft, which makes the aircraft pitch up, making the main wing destabilizing. The tail or the horizontal stabilizer are mostly situated at a larger distance from the CG, which a slight deflection of the elevator down would make the aircraft pitch down. This makes the tail a very stabilizing factor of the aircraft.

Through in our case the concept of dihedral wing is used for the front wing, and the concept of anhedral wing is used in the case of the rear wing, in order for the rear wing to eliminate the case of the downwash created by the front wing, which in case affects the rear wing, if it was placed at the same level to the front wing, is kept slightly above. About a height of 1.5 m, in order to compensate for the longitudinal and roll stability of the aircraft. The CG of the aircraft is to be kept at 5.5 meters from the front of the fuselage. These will give a good sense of control to the aircraft. The autopilot would have to perform less calculations to control the aircraft. Keeping the CG location at 5.5 meters from the fuselage provides an unstable wing at the front while a stable wing at the rear of it.

During the operation of vertical take-off the aircraft would use the all engine and use the fan control to control its engine. The moment generated by both the wings are different. The moment on the front wing is greater than the moment generated by the rear wing, which is to be stabilized by the engine control.

Considering the condition of the paramedic and the patients, the roll motion and the directional motion are done by the asymmetric moment of the flaps and also the fan speed. The turning motion in an aircraft or a VTOL is mostly characterized by the radius of the turn, the wing loading criteria and the centrifugal force acting on the aircraft. Supposed to turn left, the aircraft would deflect both its front and rear flaps on the right side of it by some angle (depending on the rate of the turn and the radius of the turn), which in turn would create asymmetric lift distribution, on both the wings. The higher lift on the right side precedes the lift on the right side and thus makes the aircraft turn.

Mostly roll or directional stability is performed by the vertical stability of the tail. But there are some other features which the conventional bomber uses that are part of a tailless aircraft. So mostly, our VTOL is based on the concept of tailless aircraft, (lacking vertical stabilizer or the yaw motion). For the use of rolling moment, some dihedral angles of 2 to 3 degrees are provided to the front wing but the main factor which aids in roll stability is the use of the engines mounted at the flaps which acts as the thrust vectoring control.

The use of the directional stability is mostly performed by the rudder (yaw motion) in an aircraft. But for a tailless aircraft, the wings are bent back at a certain angle, known as the sweep angle. Generally Sweep angles are used mostly in the fighter aircrafts, to reduce the critical mach number of the wing. Thereby reducing the local velocity over the wings and hence the aircraft could travel faster. But in the case of the stability factors, a few degrees of sweep angle provides turn stability. Hence, in our VTOL design, the rear wing is swept back by a few degrees to aid for the turn factor.

Taking about the cruise condition, where in an conventional aircraft, the flaps are retracted back to the normal position and the general elliptical lift distribution takes place over the wing. For the case of our design, similar configuration of flaps are seen and the intake of the engine produces a laminar flow over the wing, making the flow to be smooth and laminar flow above the wing. The high speed flow over the wing causes the low pressure and the high pressure created below the wing generates enough lift to carry the body.

As we have talked about the downwash created by the front wing, and the rear wing being on its wake region. So, a fine adjustment is done, where the rear wing is placed above the rear wing to neglect the downwash. A similar CFD data is analyzed below

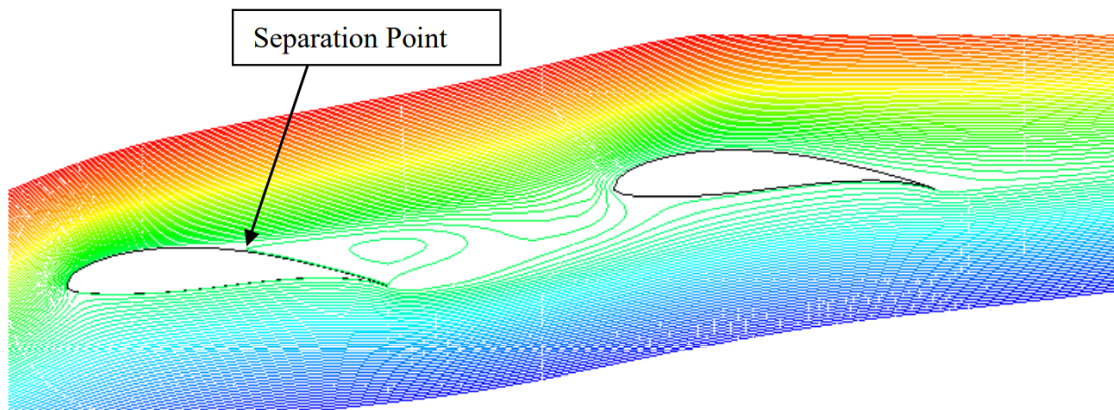


Figure 9.3 showing the Streamline at the recirculation region attached on the upstream airfoil near stall

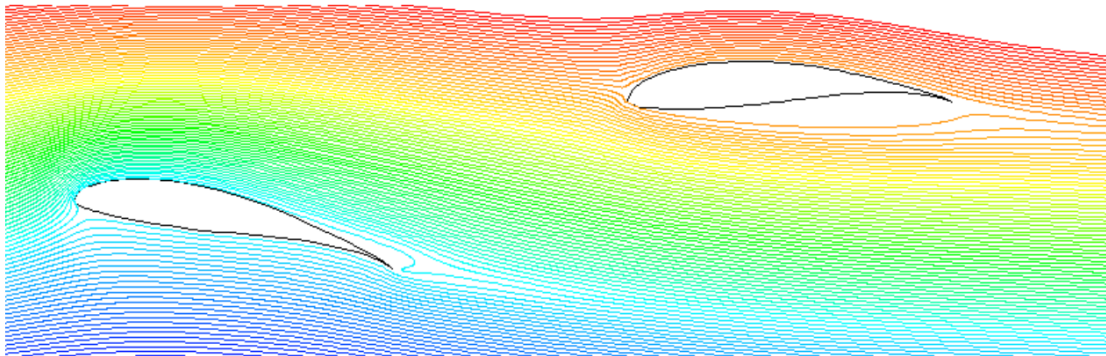
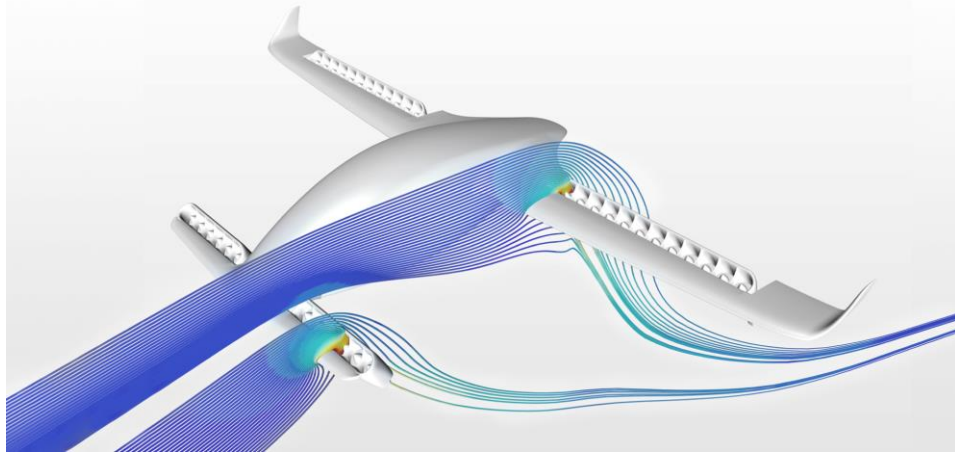


Figure 9.4 showing the velocity streamline for the tandem configuration



Figure 9.5 showing the use of the flap control for thrust vectoring



9

Figure 9.6 Cfd result depicting similar aircraft using thrust vectoring.

10. Communication System

Antenna System:

Install a high-gain directional antenna on the aircraft to establish a reliable long-range communication link. The antenna should be designed to operate in the desired frequency band and provide a focused beam to maximize signal strength.

Modulation Scheme: Choose an appropriate modulation scheme that optimizes the data rate and ensures reliable transmission over the given distance. Common choices include quadrature amplitude modulation (QAM) or orthogonal frequency-division multiplexing (OFDM).

Transceiver:

Install a transceiver system capable of transmitting and receiving data at the desired data rate. The transceiver should be compatible with the selected modulation scheme and operate within the designated frequency range.

Signal Processing:

Implement signal processing techniques to enhance the received signal quality. This may involve error correction coding, equalization, and noise reduction algorithms to mitigate channel impairments and improve data integrity.

Data Compression: Consider employing data compression techniques to

maximize the utilization of the available bandwidth. Lossless or lossy compression algorithms can be used depending on the acceptable trade-off between data size reduction and potential loss of information.

Protocol and Networking:

Implement a communication protocol stack to facilitate data transmission and establish reliable connections between the aircraft and ground stations or other aircraft. Common protocols include TCP/IP for network communication and aviation-specific protocols like ACARS (Aircraft Communications Addressing and Reporting System).

Ground Infrastructure:

Establish ground-based infrastructure such as base stations or satellite links to connect with the aircraft. These ground stations would typically have high-speed internet connections and act as the gateway between the aircraft and other communication networks.

Regulatory Considerations: Ensure compliance with regulatory guidelines and obtain necessary licenses or certifications for operating the communication system. Depending on the region, there may be specific frequency bands or power limitations to adhere to.

It's important to note that designing and implementing a complete communication system for aircraft is a complex task that involves various engineering disciplines. This overview provides a general outline, but the actual implementation would require detailed planning, testing, and expertise in aviation communication systems.

10.1 Working Principle:

Data Transmission: The communication system transmits data from the aircraft to the ground station or other aircraft. The data is first encoded and modulated into a suitable format for transmission.

Transmitter: The transmitter on the aircraft receives the encoded data and modulates it onto a carrier signal. This carrier signal is then amplified to an appropriate power level for transmission.

Antenna: The high-gain directional antenna on the aircraft transmits the modulated carrier signal, focusing it in a specific direction toward the intended receiver (e.g.,

ground station or another aircraft).

Receiver: The receiver at the ground station or receiving aircraft captures the transmitted signal using a compatible antenna.

Demodulation: The received signal is demodulated to extract the modulated data.

Signal Processing: The demodulated signal undergoes signal processing techniques to improve its quality. This may include error correction coding, equalization, and noise reduction algorithms to enhance the integrity of the received data.

Decoding: The processed signal is decoded, reverting it back to its original format.

Data Reception: The decoded data is then made available for further processing, storage, or display at the receiving end.

10.2 Line Diagram: While we can't provide a visual diagram, but can describe the basic components and their connections:

Aircraft:

- Data source (e.g., onboard computer, sensors)
- Encoder (converts data into a suitable format for transmission)
- Modulator (modulates the encoded data onto a carrier signal)
- Transmitter (amplifies the modulated signal)
- Antenna (directional antenna for transmission)

Communication Medium (air-to-ground or air-to-air):

- Transmission of the modulated signal over the desired frequency band and range

Ground Station or Receiving Aircraft:

- Receiving antenna (captures the transmitted signal)
- Receiver (captures and amplifies the received signal)
- Demodulator (extracts the modulated data from the received signal)
- Signal processing components (error correction, equalization, noise reduction)
- Decoder (reverts the modulated data back to its original format)
- Data destination or further processing/storage/display devices

Reference

1. [Microsoft Word - Abrego Dfan.doc \(nasa.gov\)](#)
2. [Ducted fans for hybrid electric propulsion of small aircraft | SpringerLink](#)
3. <https://www.rapidshift.net/electric-aircraft-calculations/>
4. <https://doi.org/10.1007/s13272-019-00398-x>
5. Determining parameters of electric power unit for light aircraft International Conference "Actual Issues of Mechanical Engineering" (AIME 2018)
6. <https://www.nrel.gov/docs/fy22osti/80220.pdf>
7. <https://www.rapidshift.net/electric-aircraft-calculations/>
8. <https://lilium.com/newsroom-detail/technology-behind-the-lilium-jet>
9. <https://link.springer.com/article/10.1007/s13272-022-00573-7>
10. <https://www.electrofluidsystems.com/Systems/Manned-Systems>
11. https://rotorcraft.arc.nasa.gov/Publications/files/Abrego2_AHS02.pdf
12. Aerodynamic Characteristics of Flow over Two Unsymmetrical Tandemly Arranged Airfoils – Numerical Simulation IOP Publishing
doi:10.1088/1757-899X/912/4/042033
13. Determination of Dispatch Takeoff Weight Dave Anderson Flight Operations Engineering Boeing Commercial Airplanes
14. [Exploring tandem wing UAS designs for operation in turbulent urban environments - Rohan Gigacz, Abdulghani Mohamed, Pakorn Poksawat, Ashim Panta, Simon Watkins, 2018 \(sagepub.com\)](#)
15. CFD ANALYSIS OF TANDEM WINGED AIRCRAFT L.Hariramakrishnan1 , K.Nehru1 , V.Rajashree2 , S.Gowtham3 , R.Karthika4 1,2Assistant Professor, Department of Aeronautical Engineering 1,2,3,4 SNS College of Technology, Coimbatore